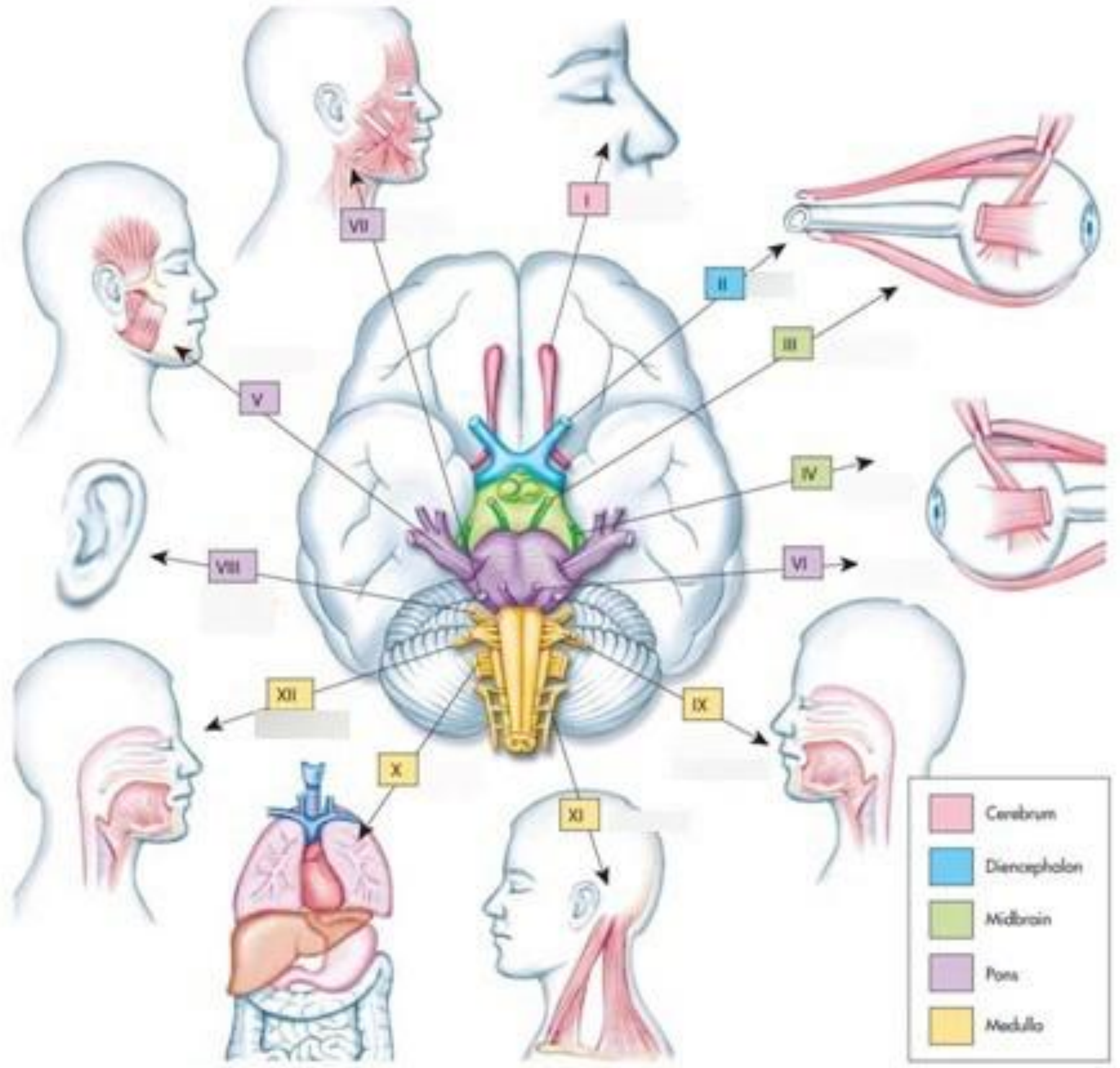
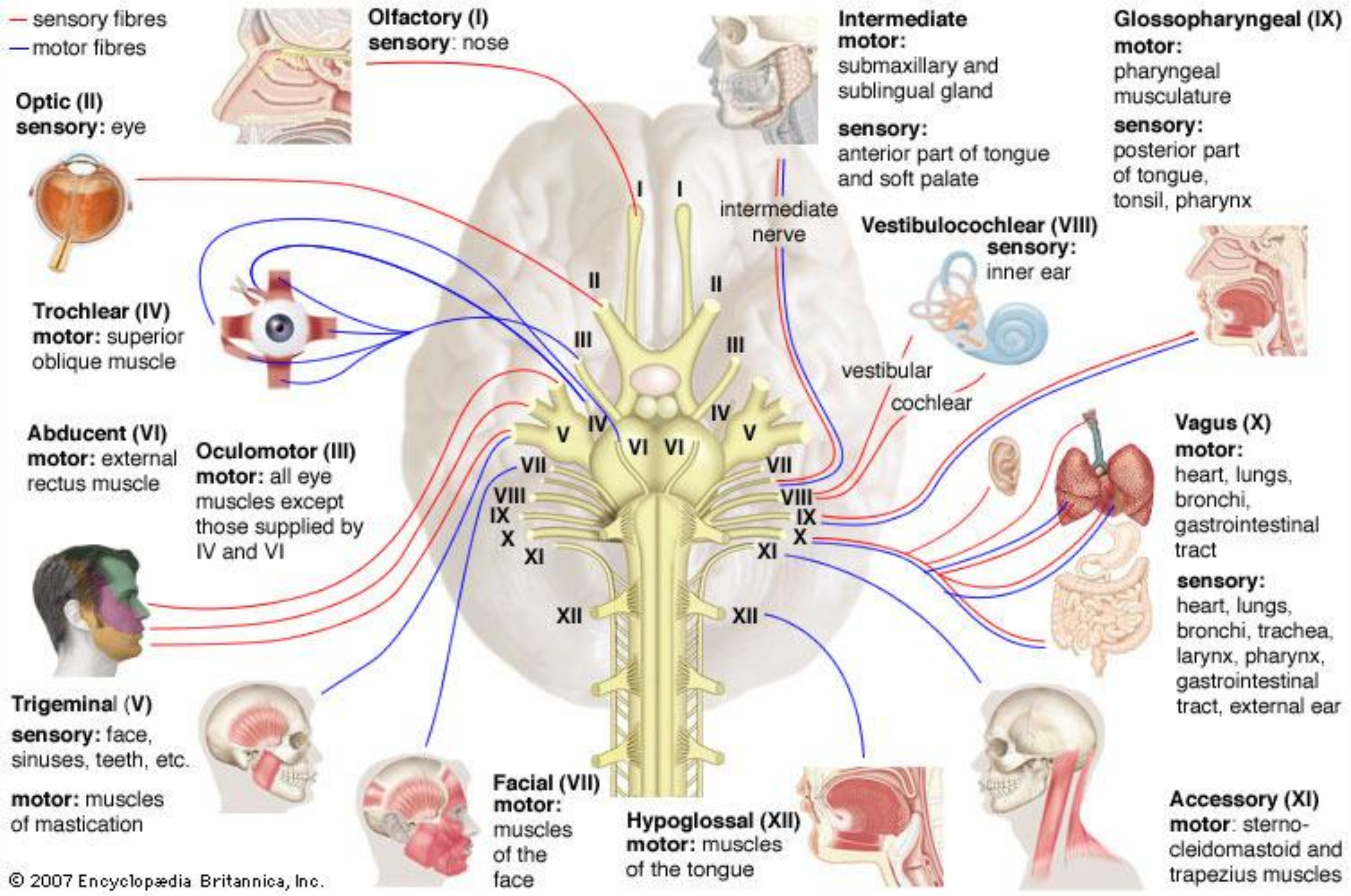
OSTEOPATHIC APPROACH TO CRANIAL NERVES

STACIA SLOANE, DO





sensory fibres
motor fibres
Olfactory (I)
sensory: nose
Optic (II)
sensory: eye
Trochlear (IV)
motor: superior oblique muscle
Abducent (VI)
motor: external rectus muscle
Oculomotor (III)
motor: all eye muscles except those supplied by IV and VI
Trigeminal (V)
sensory: face, sinuses, teeth, etc.
motor: muscles of mastication
Facial (VII)
motor: muscles of the face
Hypoglossal (XII)
motor: muscles of the tongue
Intermediate
motor: submaxillary and sublingual gland
sensory: anterior part of tongue and soft palate
intermediate nerve
Vestibulocochlear (VIII)
sensory: inner ear
vestibular
cochlear
Glossopharyngeal (IX)
motor: pharyngeal musculature
sensory: posterior part of tongue, tonsil, pharynx
Vagus (X)
motor: heart, lungs, bronchi, gastrointestinal tract
sensory: heart, lungs, bronchi, trachea, larynx, pharynx, gastrointestinal tract, external ear
Accessory (XI)
motor: sterno-cleidomastoid and trapezius muscles
I I II II III III IV IV V V VI VI VII VII VIII VIII IX IX X X XI XI XII XII

OBJECTIVES

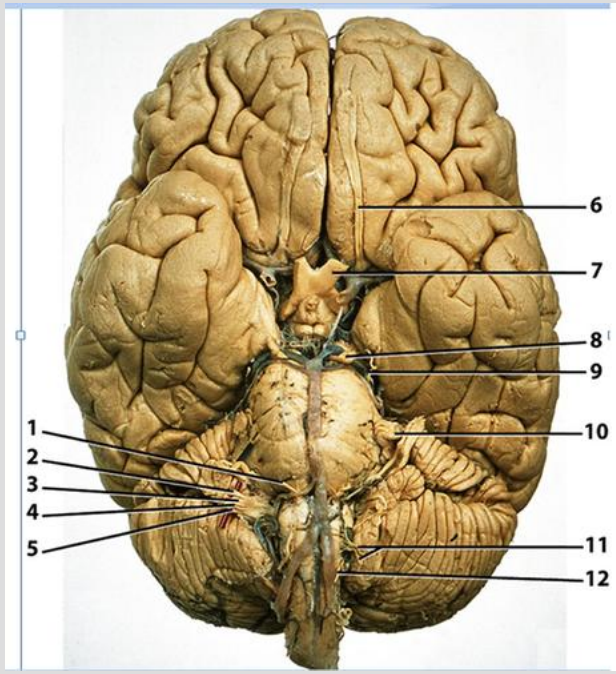
Identify	Identify the twelve (12) cranial nerves by name, emphasized function, and pathway
Identify	Identify the shared cranial nerve pathways involving the CN V, VII, IX
Identify	Identify the cranial anatomy and foraminal associations for each cranial nerve and vascular component as emphasized
Identify	Identify the neurovascular structures that pass through the Cavernous Sinus
Identify	Identify the neurovascular structures that pass through the Jugular Foramen
Identify	Identify the neurovascular structures that pass through the Superior Orbital Fissure
Identify	Identify cranial nerves that carry parasympathetic fibers, define the nerves, ganglia, foramen and paths of the involved nerves.
Identify	Identify possible clinical presentation that involve cranial nerve(s) and the possible anatomical site of neurovascular compromise or dysfunction

OSTEOPATHIC CONSIDERATIONS

- The cranium is **alive and compliant**, and as such has intrinsic motion and physiology that represents reciprocally interrelated **structure and function**
- With the exception of oral/sinus spaces and enclosures, there is **no empty free space** in the cranium that is not filled with living vibrant tissue
- Cranial nerves are directly influenced by the surrounding fluids, membranes, and the motion of the Primary Respiratory Mechanism
- They are subject to dysfunction in the surrounding tissues

THINGS TO KNOW

- Emerge from the brain superior to foramen magnum
- Cranial nerve sensory and motor fibers enter and exit the brainstem at the **same location** (contrast spinal nerves where the motor efferent fibers exit at the anterior horn while sensory afferent fibers enter at the dorsal root)
- There are many *shared* pathways
- Cranial Nerves have seven (7) different functional conduction type:
  - General Somatic Afferents
  - General Visceral Afferents
  - General Visceral Efferents
  - General Somatic Efferents
  - Special Somatic Afferents
  - Special Visceral Afferents
  - Special Visceral Efferents



General somatic afferents (somatic sensation):
→ E.g., fibers convey impulses from the skin and striated muscle spindles

General visceral afferents (visceral sensation):
→ E.g., fibers convey impulses from the viscera and blood vessels

General visceral efferents (visceromotor function):
→ Fibers innervate the smooth muscle of the viscera, intraocular muscles, heart, salivary glands, etc.

General somatic efferents (somatomotor function):
→ Fibers innervate striated muscles

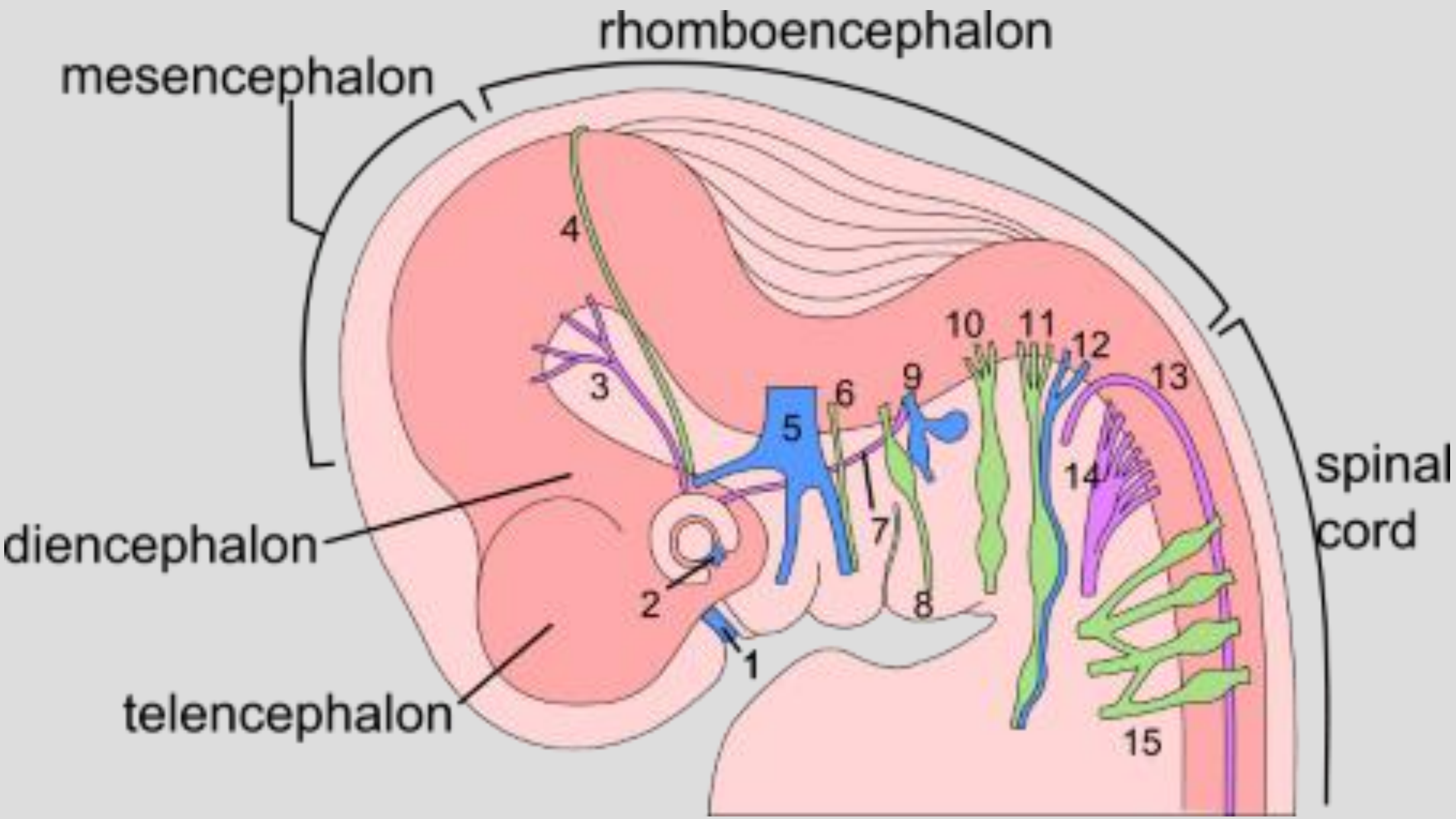
Additionally, cranial nerves may contain special fiber types that are associated with particular structures in the head:

Special somatic afferents:
→ E.g., fibers conduct impulses from the retina and from the auditory and vestibular apparatus

Special visceral afferents:
→ E.g., fibers conduct impulses from the taste buds of the tongue and from the olfactory mucosa

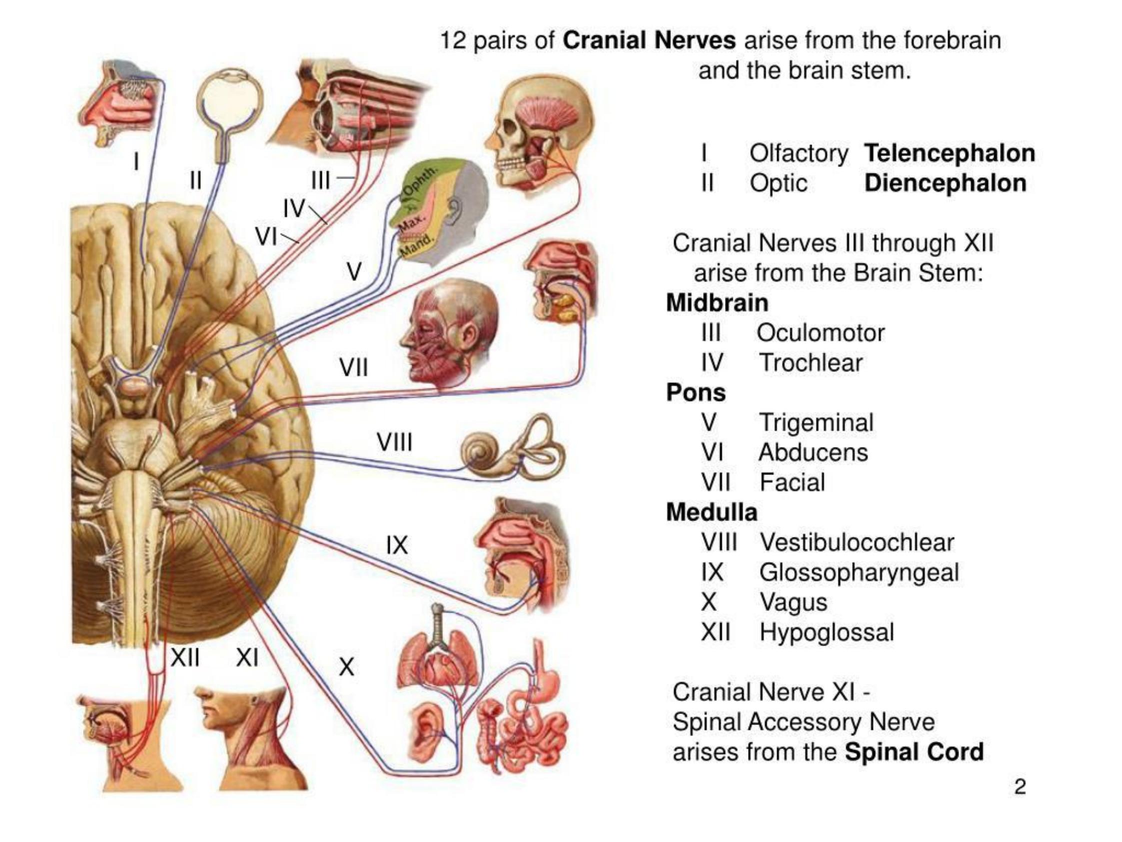
Special visceral efferents:
→ E.g., fibers innervate striated muscles derived from the branchial arches *(branchiogenic efferents and branchiogenic muscles)*

FORMING CRANIAL NERVES

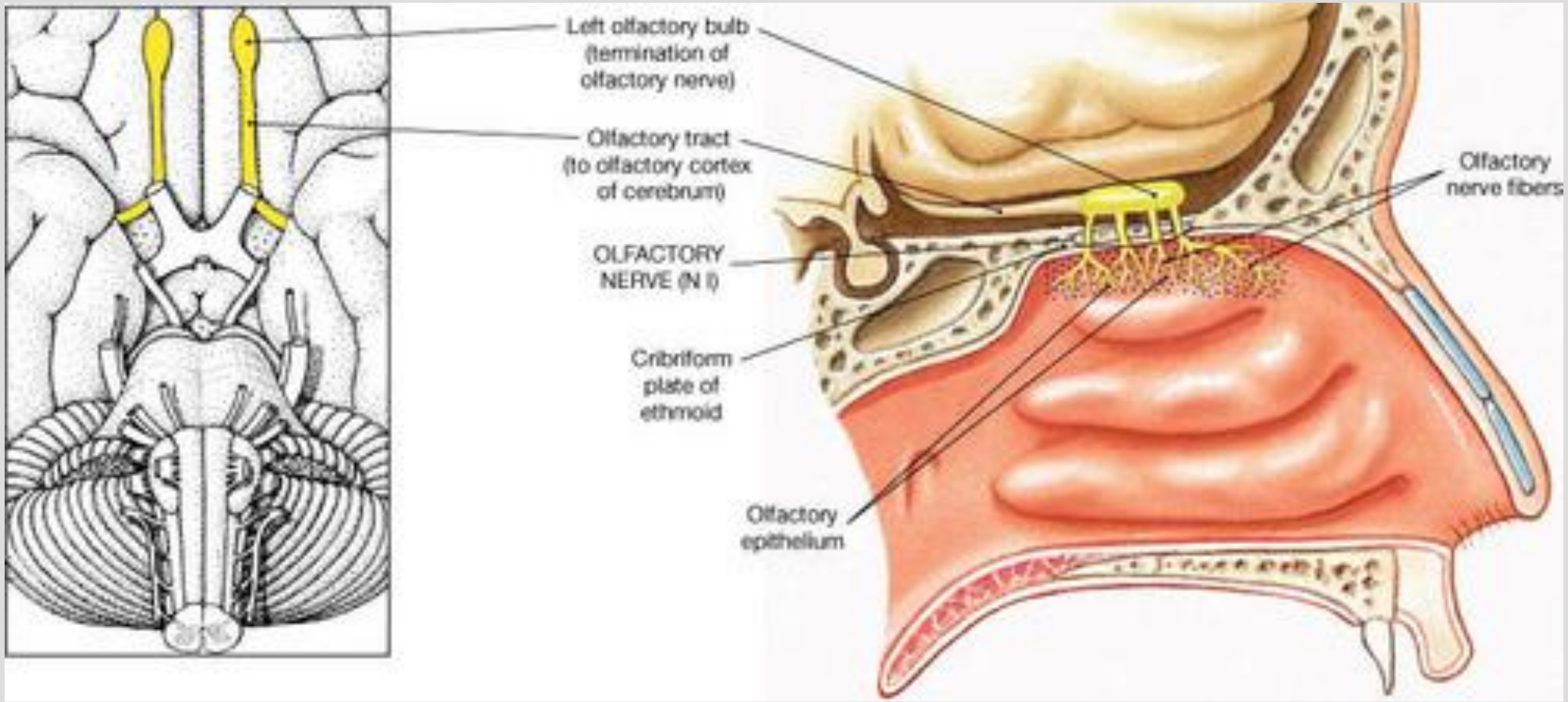


CRANIAL NERVE DYSFUNCTION

- Cranial Nerve Dysfunction may have multifocal causation
- Common processes leading to dysfunction:
  - Metabolic
  - Genetic
  - Teratogenic
  - Vascular
  - Inflammatory
  - Trauma and/or Increased intracranial pressure
- Often some IMPINGEMENT compromises axonal flow leading to neural dysfunction
  - Membranous
  - Articular
  - Fluidic



CN I: OLFACTORY NERVE



CN I: OLFACTORY NERVE

Origin:

- Telencephalon Brain Region

Fiber Types:

- Special Visceral Afferents (smell)

Path:

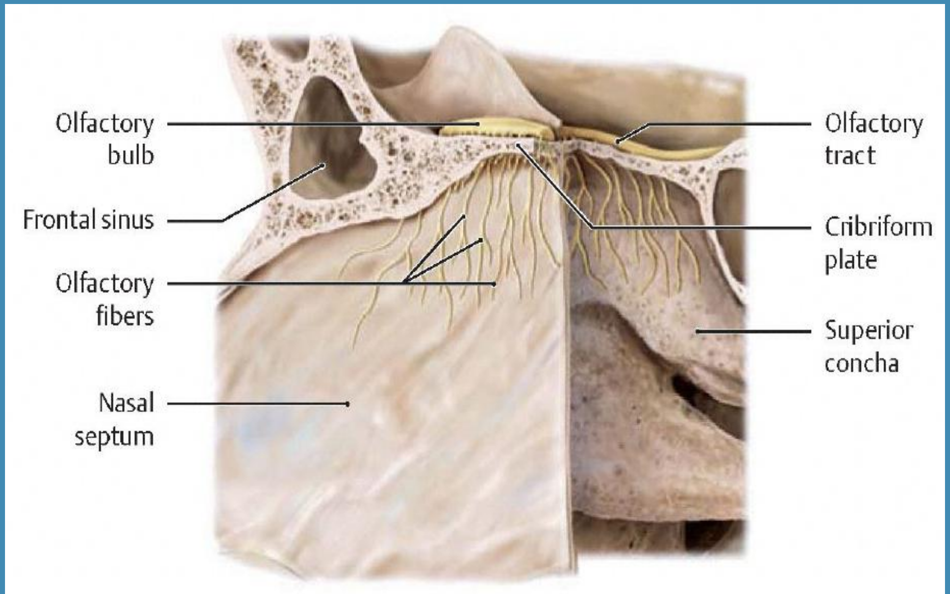
- Unmyelinated axons from the mucosa of the superior aspect of the nasal cavity are bundled as the olfactory nerve.
- They then pass through the *Ethmoid cribiform plate* into the *anterior cranial fossa* to synapse within the olfactory bulbs.
- Neural flow continues through the olfactory tracts overlying the *Sphenoid crest and olfactory grooves* to end in the cerebral cortex, amygdala and other areas.

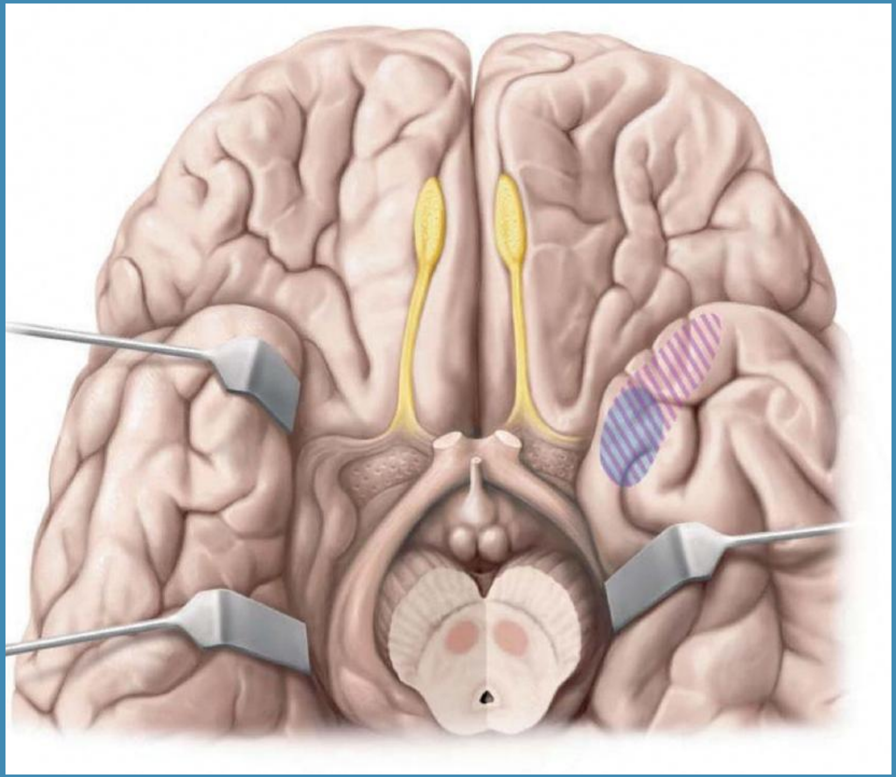
Clinical/Osteopathic Considerations:

- *Disruption of the Frontal notch (Frontal bones), Sphenoid Lesser wings, Ethmoid/cribiform plate as in trauma*
- *Dysosmia or Anosmia*

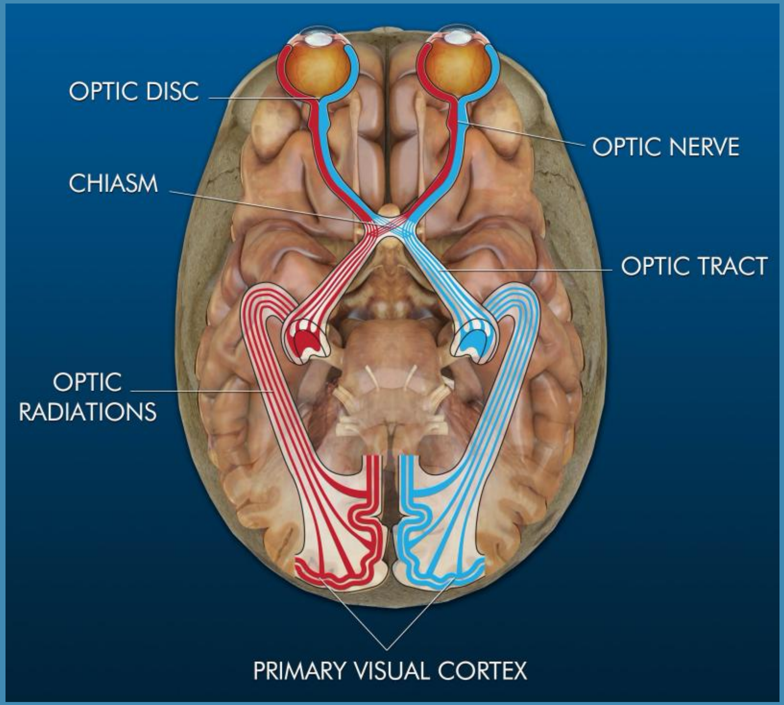
** One of two CNs that are true CNS extensions covered by meninges

OLFACTORY NERVE





CNII: OPTIC NERVE



Origin:

- Diencephalon Brain Region

Fiber Types:

- Special Somatic Afferents (Binocular Vision)

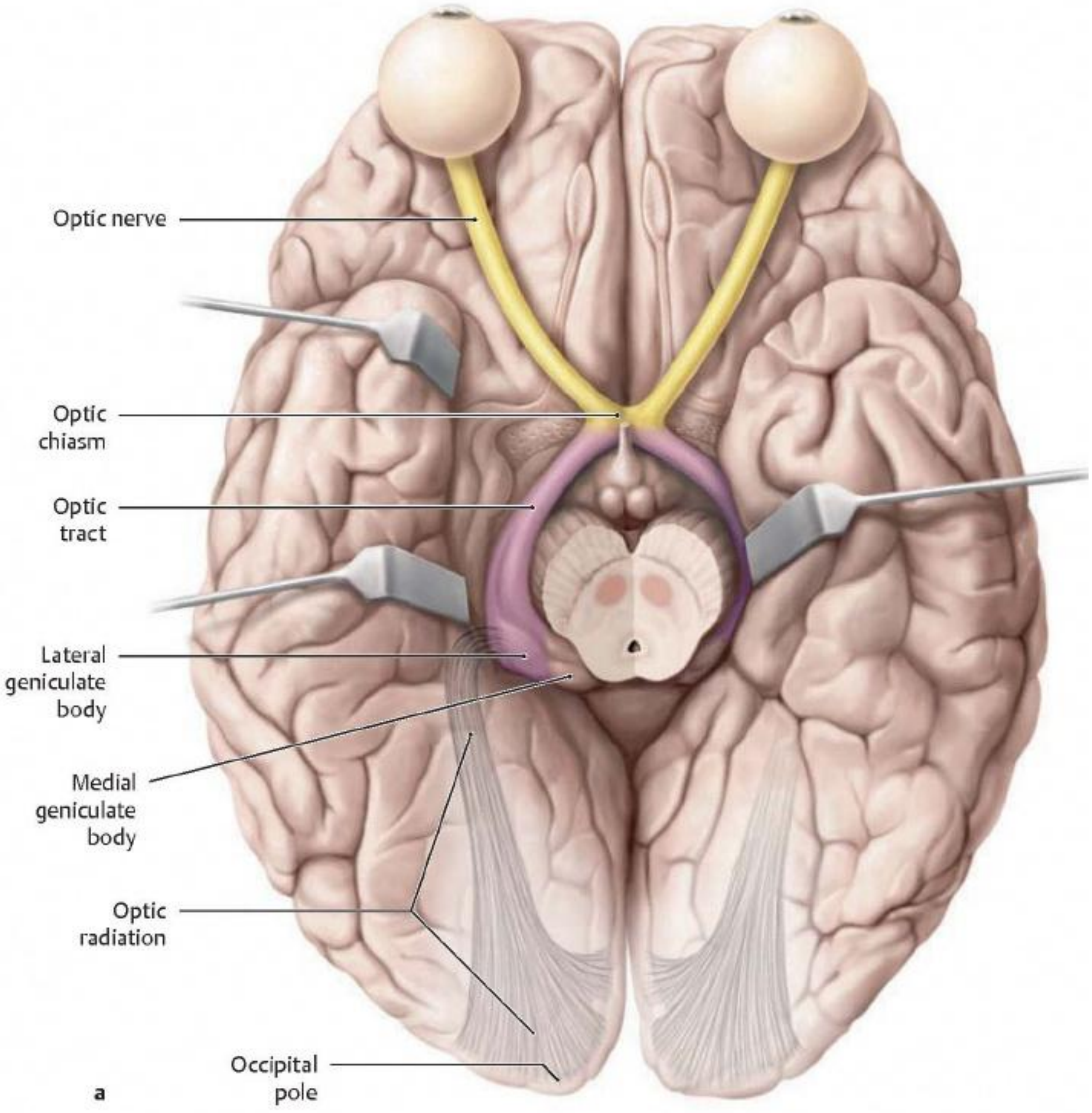
Path:

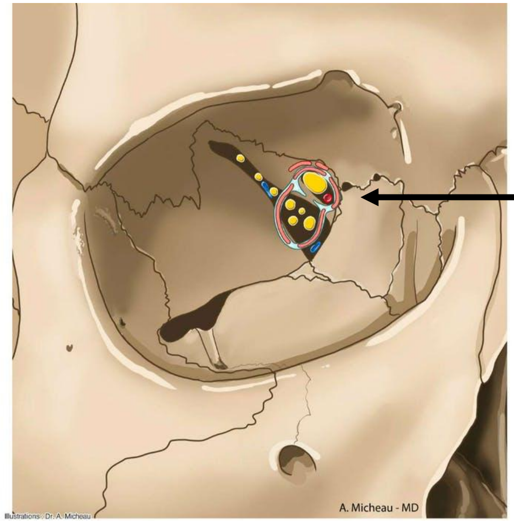
- Emerges from the retinal portion of the eye, through the *common tendinous ring (Ring of Zinn)*, into the *middle cranial fossa* by way of the Sphenoid *optic canal*
- From there it proceeds to the *Sphenoid optic chiasm* where some of its neural flow will traverse to the contralateral side to the optic tract to synapse on the lateral geniculate ganglion, while others will remain ipsilateral to its respective geniculate ganglion

Clinical/Osteopathic Considerations:

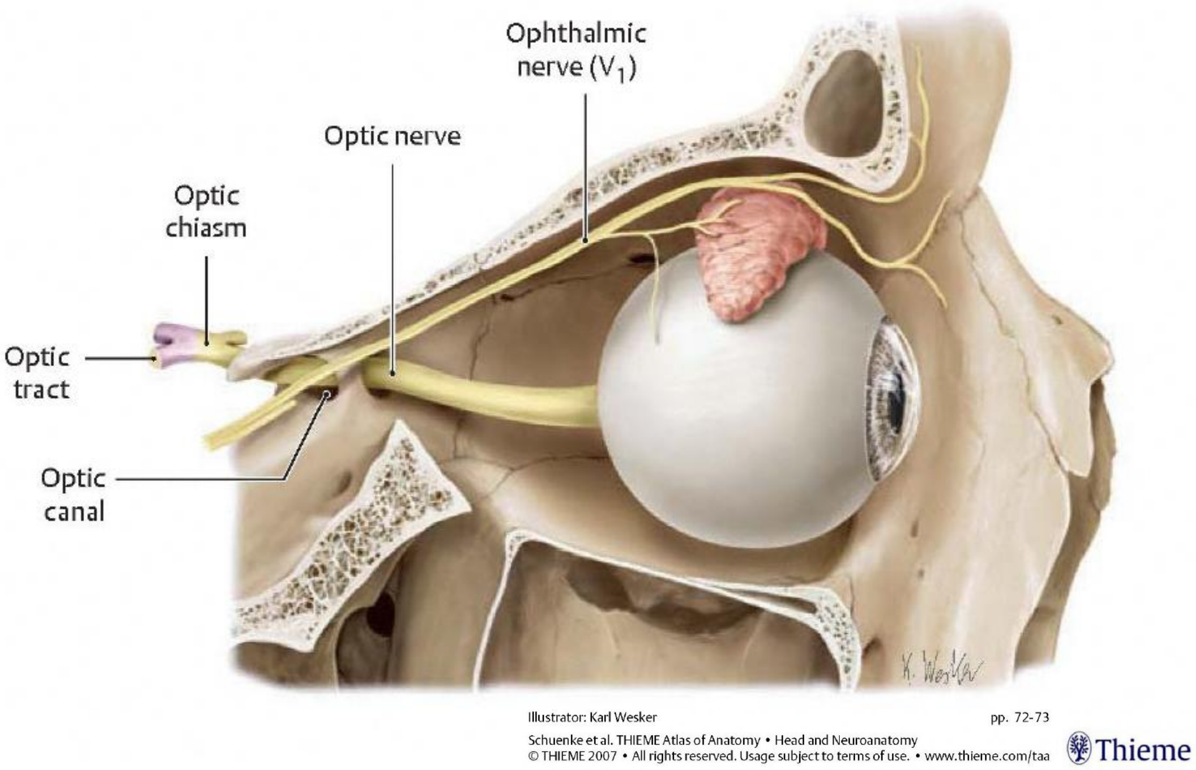
- *Developmental issues* of the *Sphenoid Lesser Wings* to the *Sphenoid Body* in forming the *optic foramen* may be of concern in birth trauma
- *Greater and Lesser wing* association as the attachments to the *common tendonous ring*
- *Vascular/lymphatic drainage* from the orbit has a component that relies on the PRM. This may have consequences on the vitality and axonal health of the nerves and tissues that reside within the orbit

** One of two CNs that are true CNS extensions covered by meninges

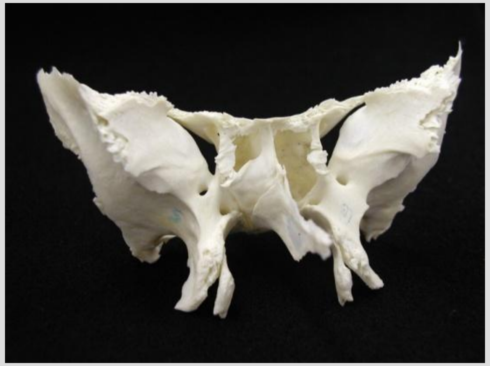


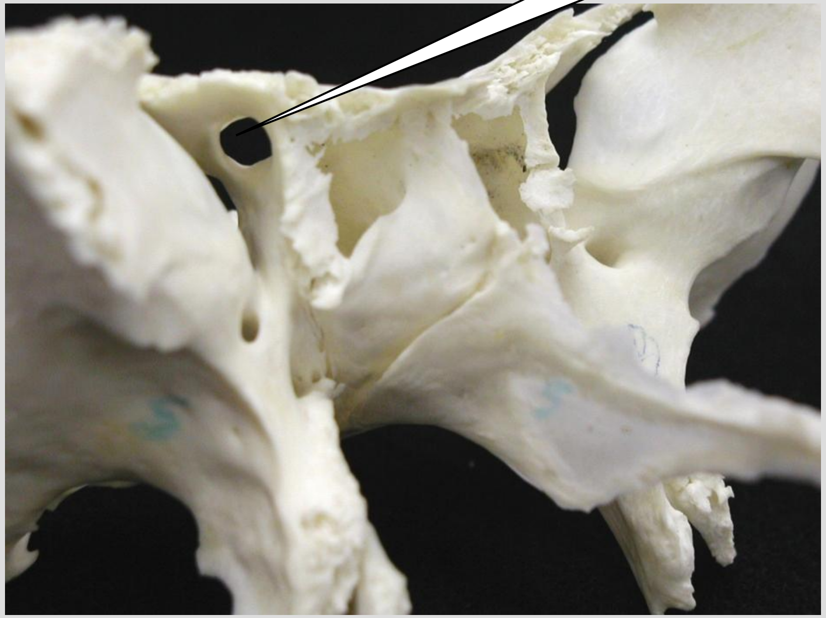


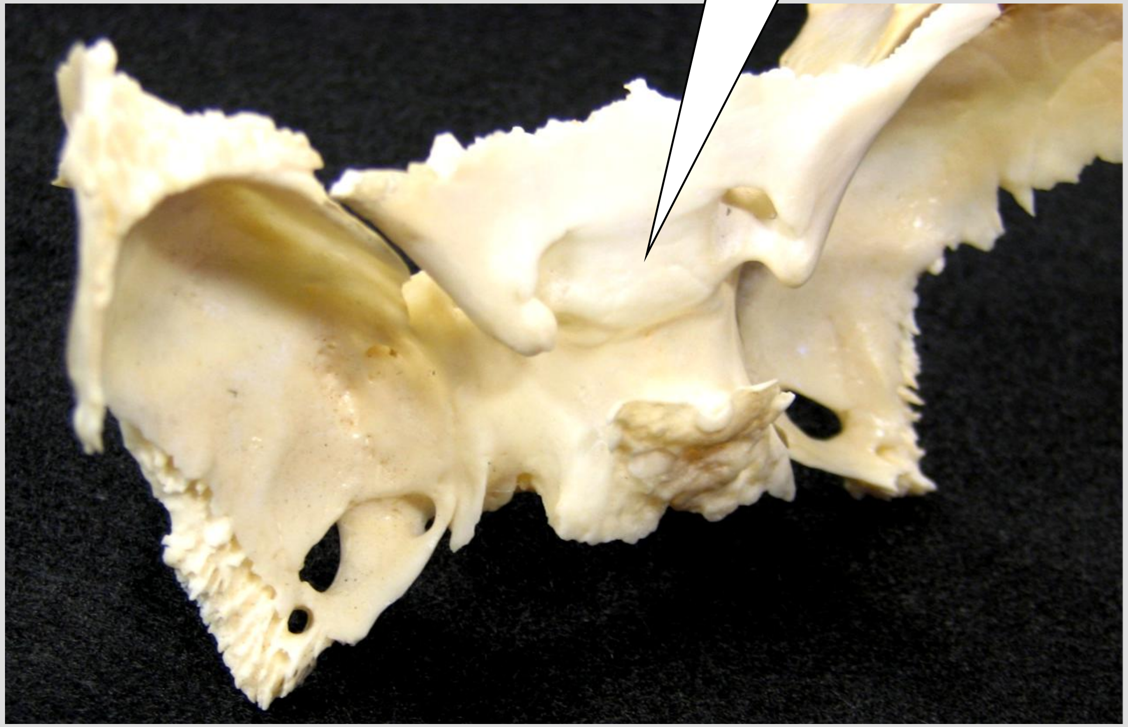
Common Tendinous Ring



OPTIC PATHWAY







OPTIC CANAL AT BIRTH

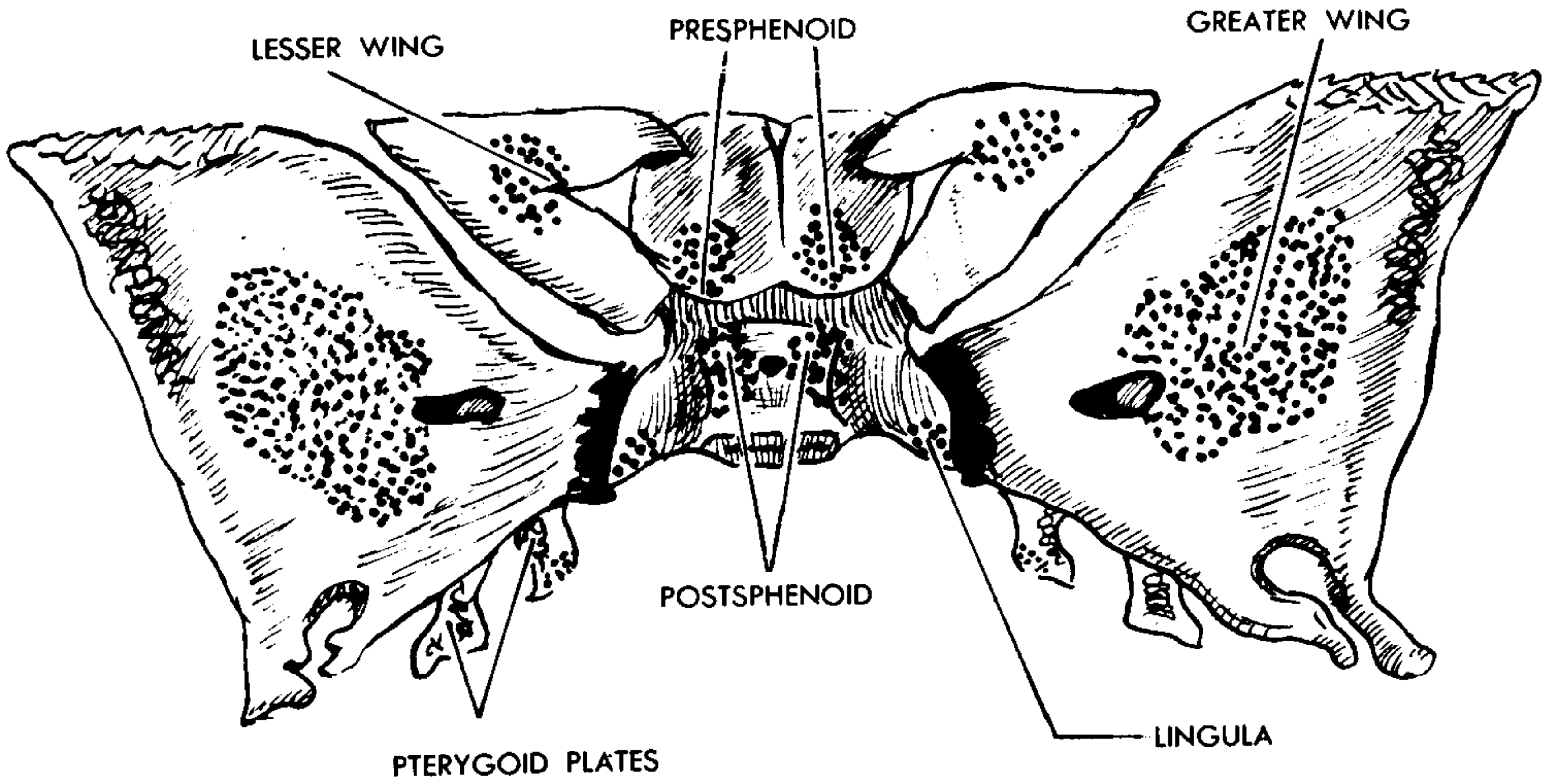
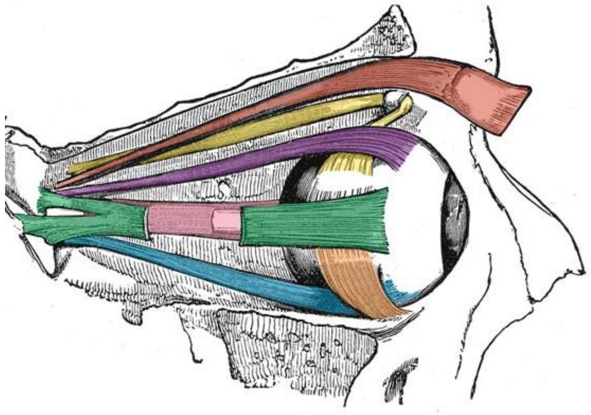
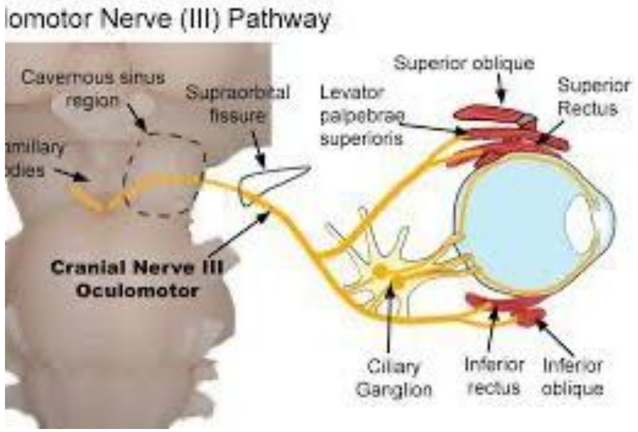


FIG. 21. SPHENOID BONE AT BIRTH.





CNIII: OCULOMOTOR NERVE

Origin:

Mesencephalon (Midbrain)

Oculomotor Nucleus

Visceral Oculomotor Nucleus (Edinger-Westphal)

Fiber Types (2):

Somatic Efferent: Extraocular Muscles (EOM)- Superior, Medial and Inferior Recti and Inferior Oblique. Levator Palpebrae Superioris (opens eye lid).

Visceral Efferent (parasympathetic preganglionic to ciliary ganglia): Pupillary Constrictor and Ciliary Muscle (lens accommodation)

Path:

The somatic portion proceeds anteriorly from its nucleus through the *cavernous sinus, superior orbital fissure, and common tendinous ring* to synapse on the EOMs.

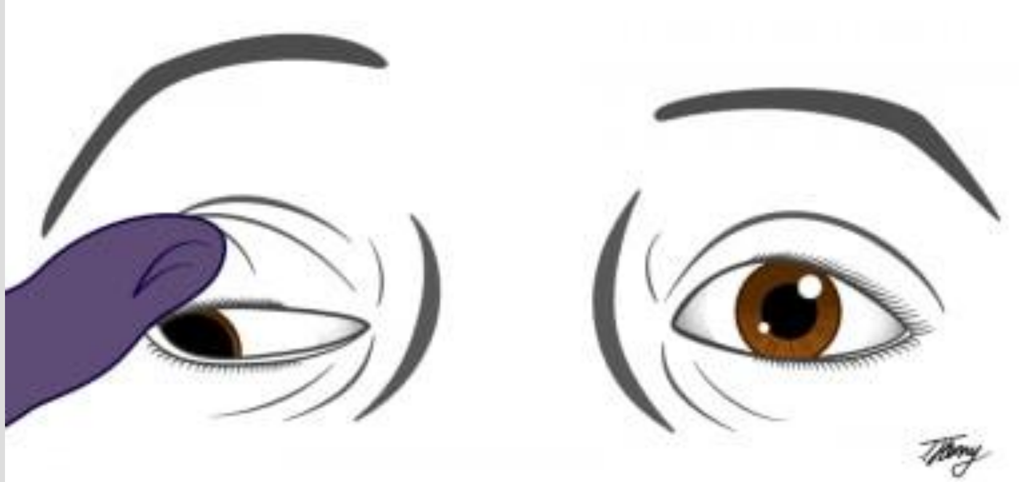
The visceral portion supplies preganglionic parasympathetic efferent innervation distal of the *common tendinous ring* to synapse on the ciliary ganglion innervating the pupillary sphincter and ciliary muscles of the eye

Clinical/Osteopathic Considerations:

Superior orbital fissure formed by the sphenoid greater and lesser wing, the state of the cavernous sinus.

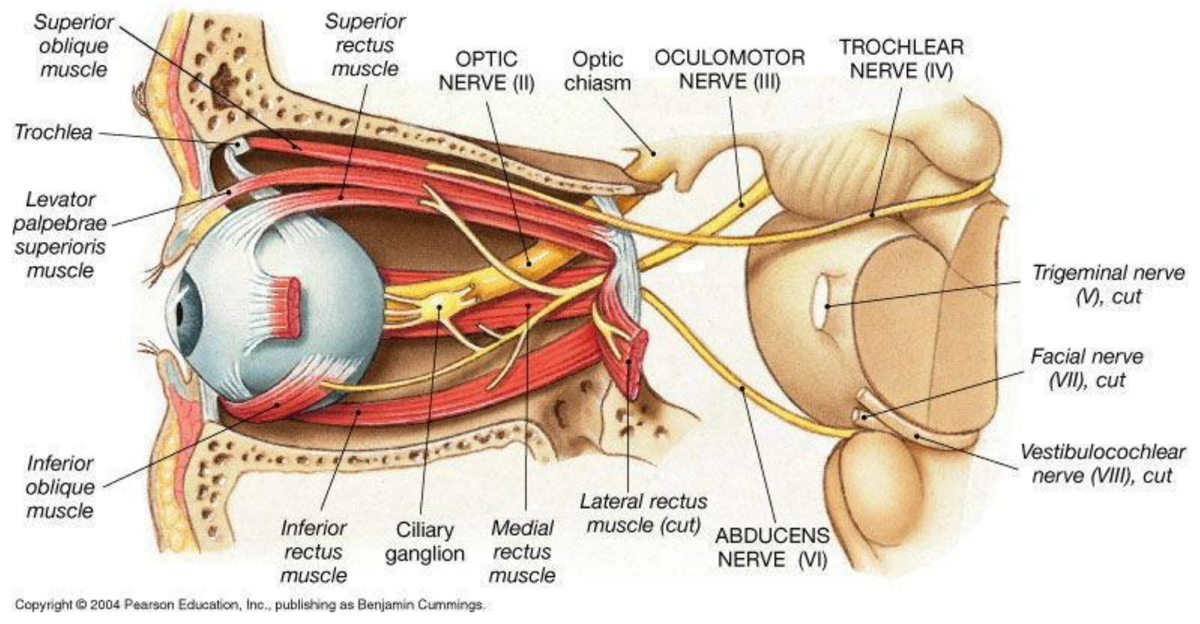
Symptoms include external strabismus, ptosis, mydriasis (pupillary dilatation)

OCULOMOTOR DAMAGE



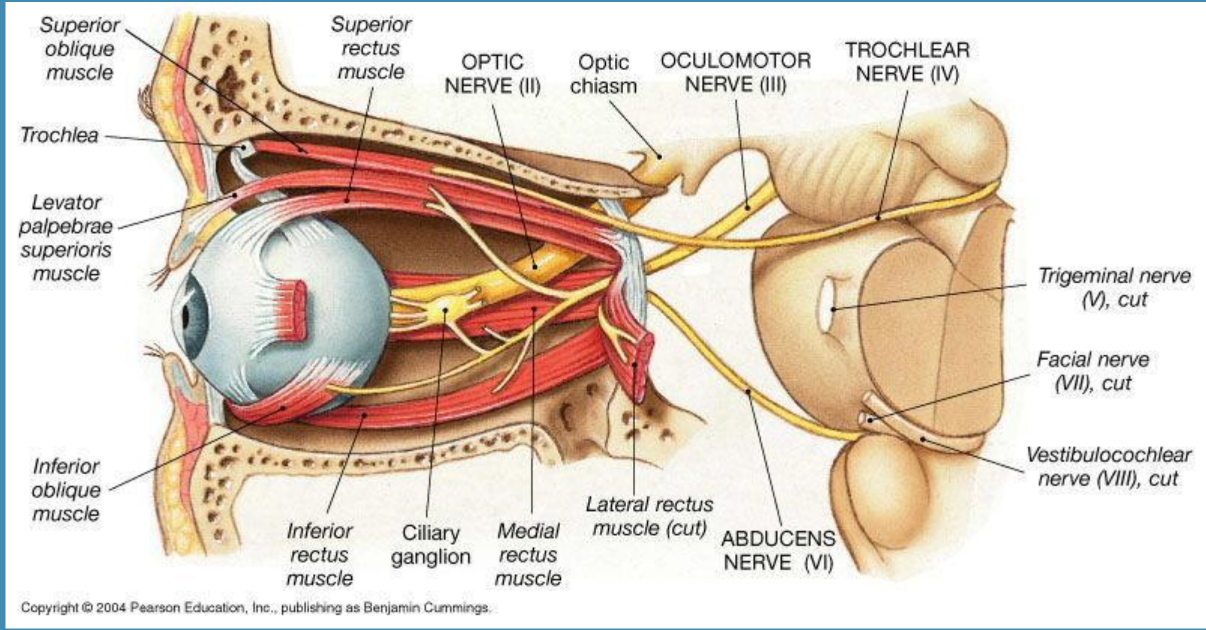
- Illustration of a complete right oculomotor palsy demonstrating the classic "down and out" appearance, complete ptosis and mydriasis of the right eye. Courtesy of Tyler Henry, MD, Medical Illustrator (tylerhenrymd.com).

CN IV: TROCHLEAR NERVE



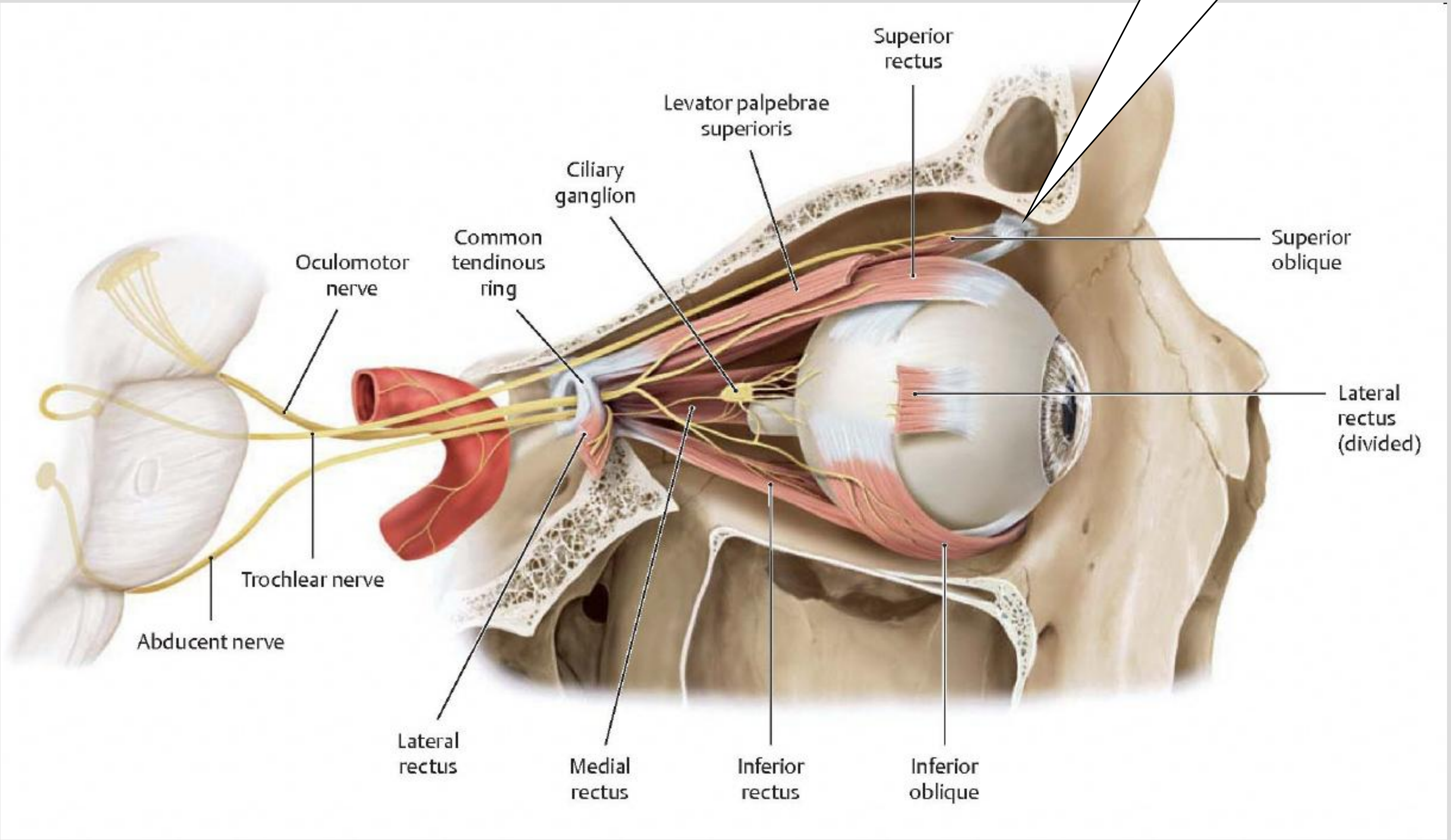
- Origin:
  - Mesencephalon (posterior aspect of the midbrain)
  - Nucleus of the Trochlear Nerve
- Fiber Types:
  - Somatic Efferent: Superior Oblique
- Path:
  - Emerges from the posterior aspect of the midbrain
  - **Crosses** to the contralateral side then courses laterally and anteriorly around the cerebral peduncles,
  - Pierces dura to enter the **cavernous sinus**, on to the orbit via the superior orbital fissure **avoiding the common tendinous ring** to synapse on the EOM
  - Note attachment onto Frontal Bone
- Clinical/Osteopathic Considerations:
  - Its anatomical approximation to the dural thickenings (plica) of the tentorium cerebelli free border and its crossing over the tentorium cerebelli attached border lends for a possible site of constriction as well as the functional state of the cavernous sinus & superior orbital fissure
  - Palsy Symptoms include: **difficult downward and outward movements** of the affected eye

CN VI: ABDUCENT NERVE



- Origin:
  - Midbrain (Pons)
  - Nucleus of Abducent Nerve (some interconnections with the oculomotor nucleus)
- Fiber Type:
  - Somatic Efferent: Lateral Rectus
- Path:
  - *Emerges from its nucleus and pierces the dura just inferior to the tentorium cerebelli attached boarder that forms the petrosphenoid ligament .*
  - *It continues on through the cavernous sinus, superior orbital fissure, and common tendonous ring, to the synapse on it's EOM*
- Clinical/Osteopathic Considerations:
  - CN VI neuropathy is common with dental trauma or dental extraction strains imparted to the dura specifically at the petrosphenoid ligament
  - Symptoms of lateral as well as medial rectus dyscoordination can cause diplopia, strabismus, & nystagmus

Trochlea sling



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pp. 72-73

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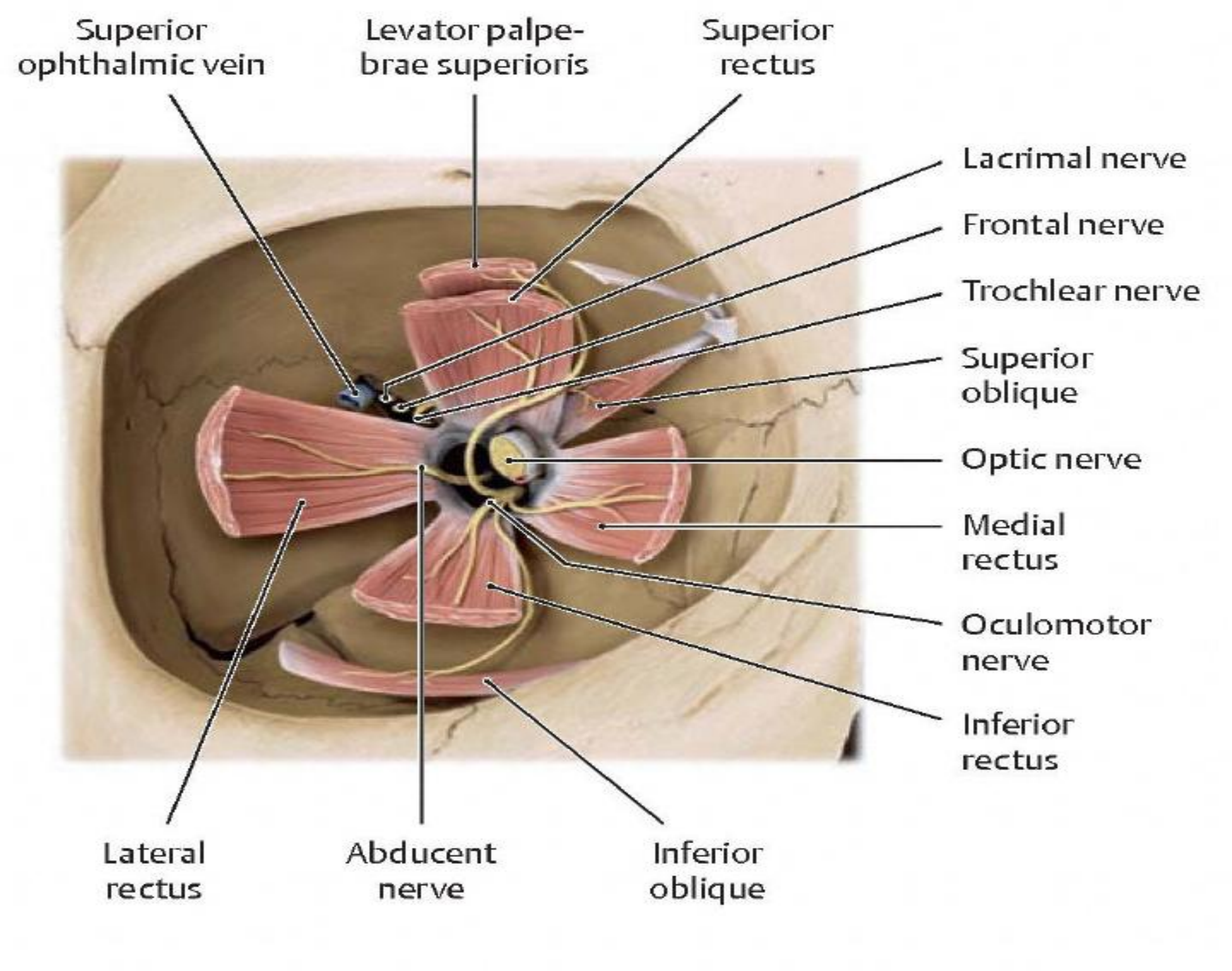

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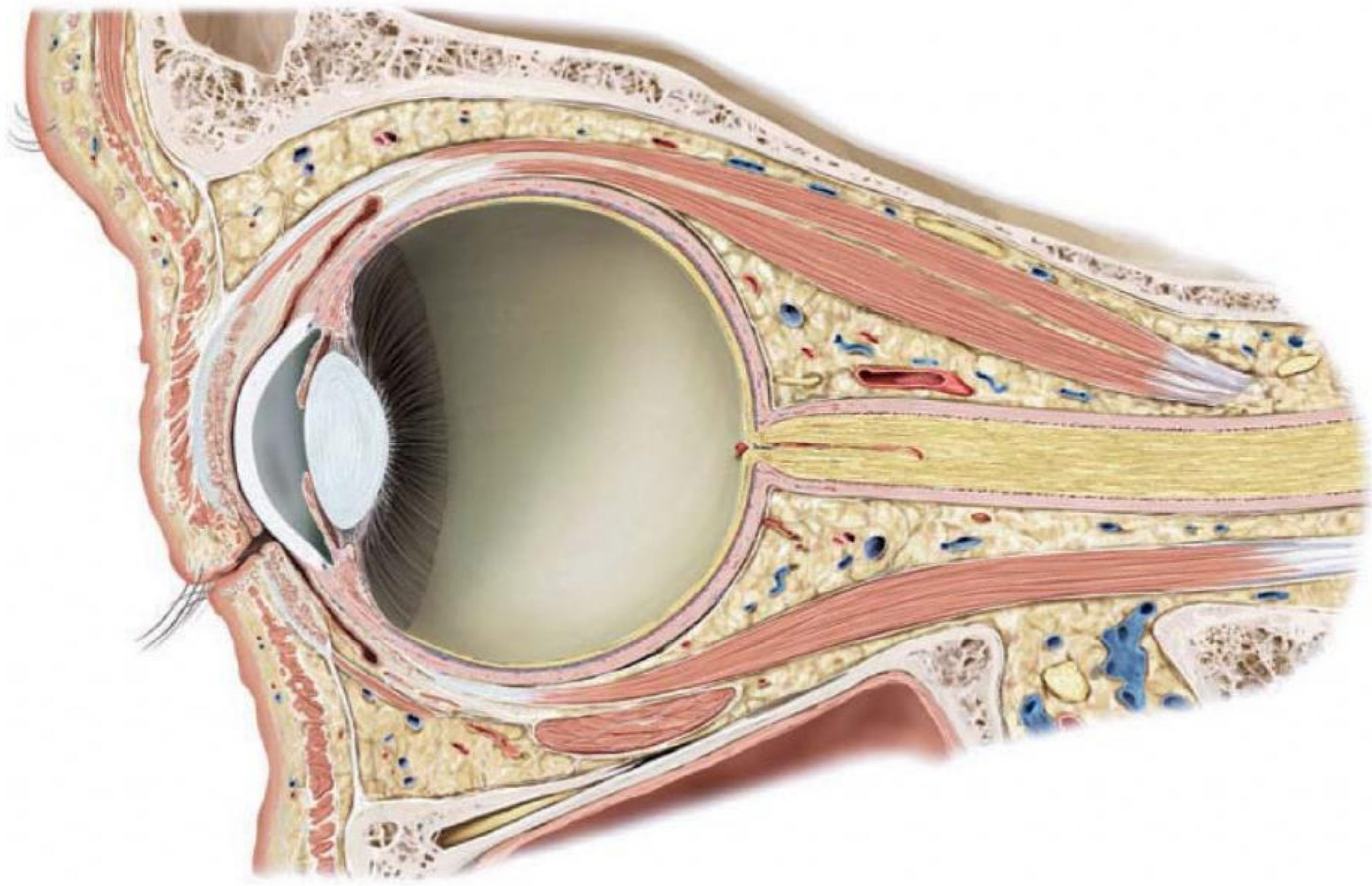

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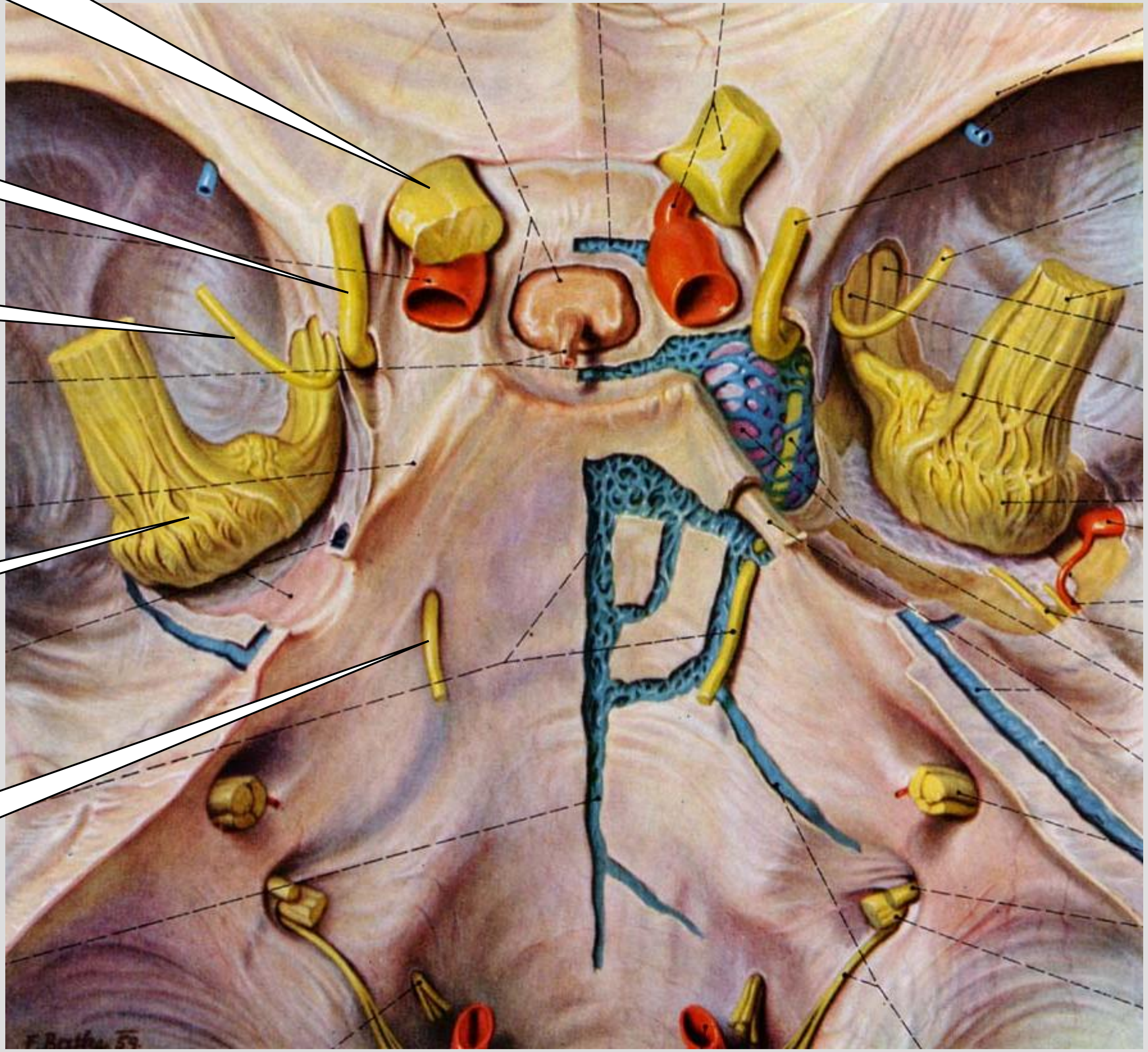
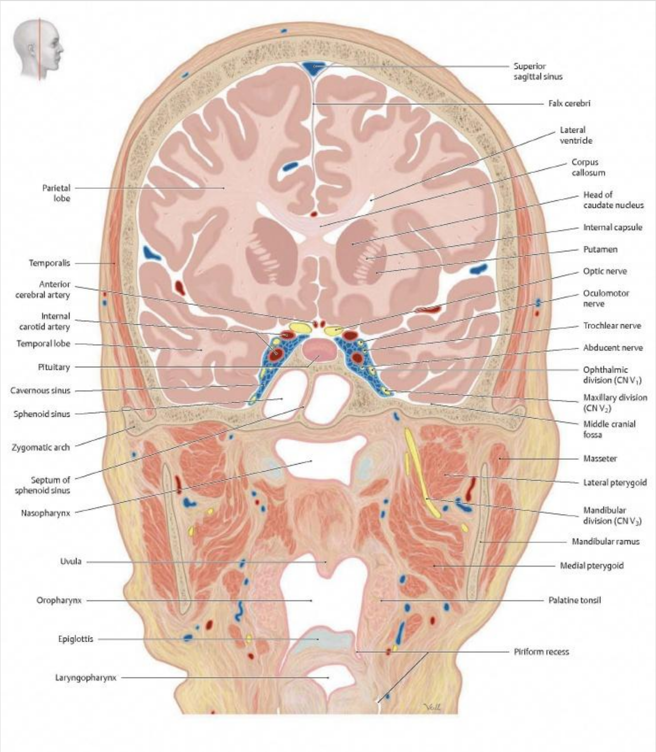
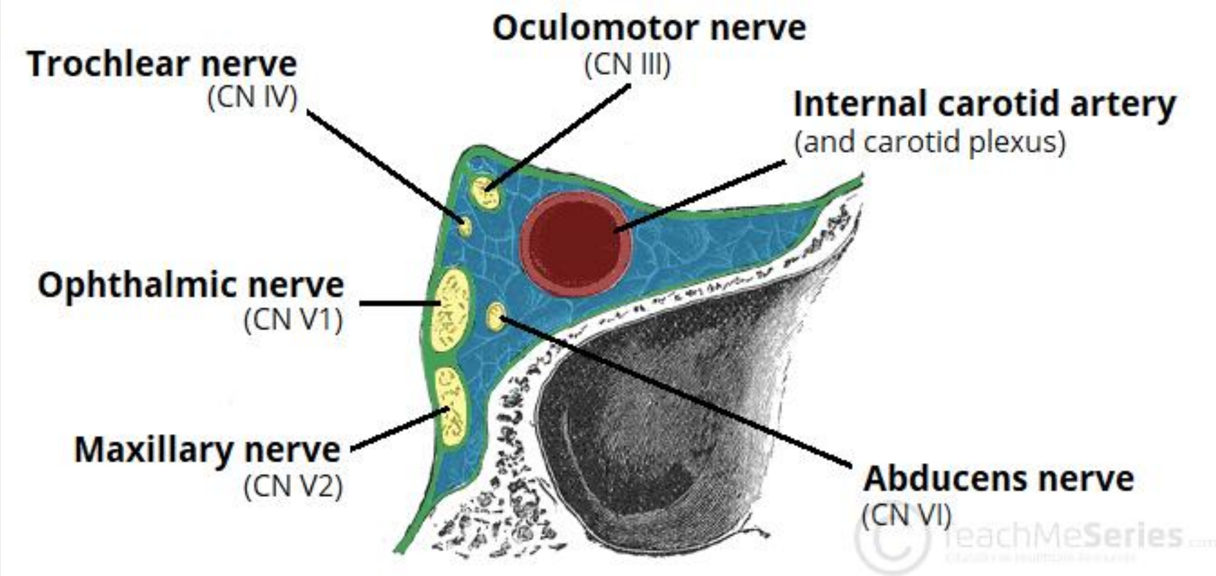
CRANIAL BASE– SUPERIOR VIEW



CAVERNOUS SINUS





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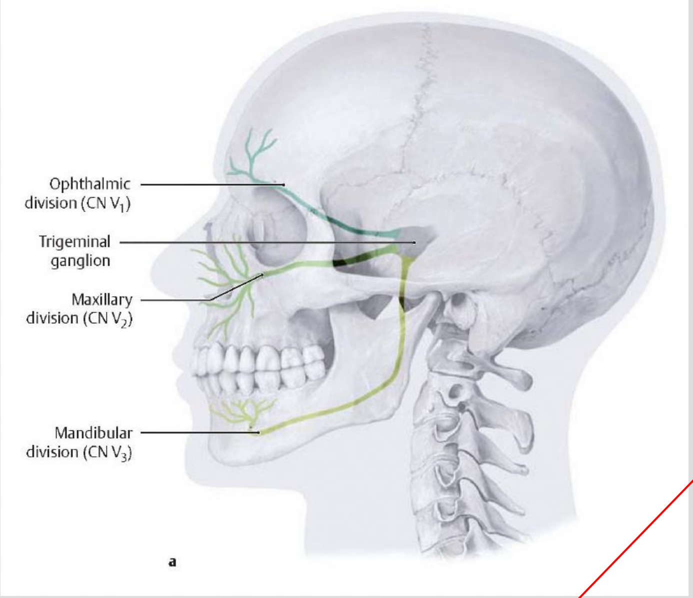
CNV: TRIGEMINAL NERVE

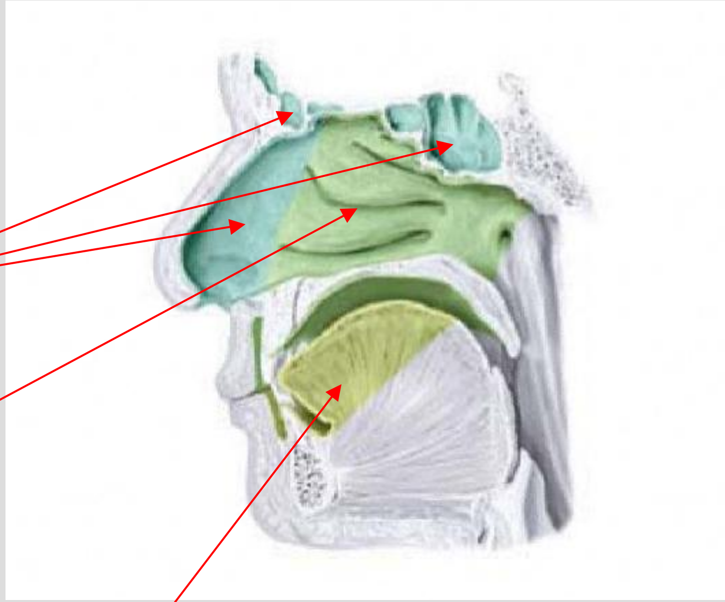
- Central Origins or Targets:
  - Trigeminal mesencephalic nucleus
    - Proprioception from mastication muscles
  - Principle (pontine) nucleus
    - Mediates touch sensation
  - Trigeminal spinal nucleus
    - Mediates pain, temp, and touch sensations
  - Trigeminal motor nucleus
    - *Motor for mastication*

- Utilizes *three divisions* from the Trigeminal (Gasserian) ganglion
  - *Ophthalmic* (V_1)
  - *Maxillary* (V_2)
  - *Mandibular* (V_3)

CNV: TRIGEMINAL NERVE

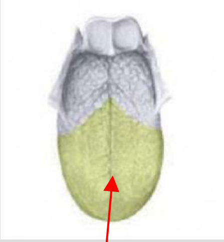
- Fiber Types
  - Special Visceral Efferent *(V_3)*
    - *Muscles of mastication– temporalis, masseter, medial & lateral pterygoid muscles*
    - Oral floor muscles – *mylohyoid, anterior belly of the digastric*
    - Middle ear muscles – *tensor tympani (Ossicle transmission attenuator)*
    - Pharyngeal muscles – tensor veli palatini
  - General and Special Somatic Afferent
    - *Sensory from the head and face, nasopharyngeal mucosa, anterior 2/3s of the tongue (V_1,V_2,V_3)*
      - Gustatory fibers of the *chorde tympani* nerve (Facial - CN VII) utilize lingual nerve branches of V_3
  - General Visceral Efferent
    - Visceral efferent from other cranial nerves share the trigeminal distribution:
      - Parasympathetic from the Facial nerve (CN VII) utilizes
        - V_2 (lacrimal nerve) to innervate the lacrimal gland
        - V_2 (zygomatic nerve)
        - V_3 (lingual nerve & chorde tympani nerve) to innervate the submandibular & sublingual glands
      - Parasympathetic innervation to the parotid gland from the Glossopharyngeal (CN IX) utilize V_3 (auriculotemporal nerve)

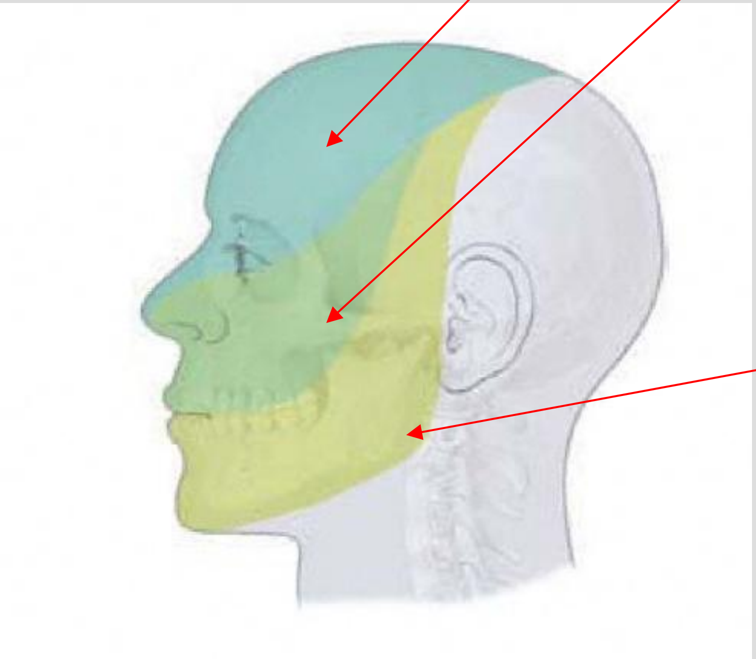




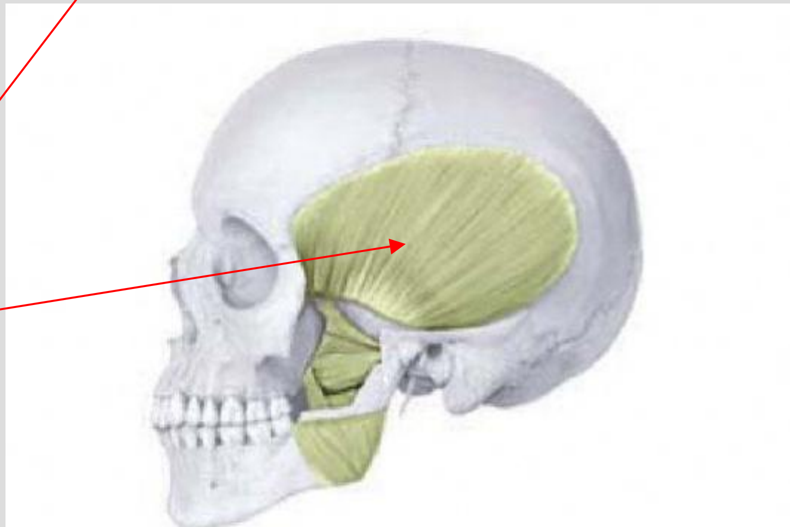
V1

V2





V3



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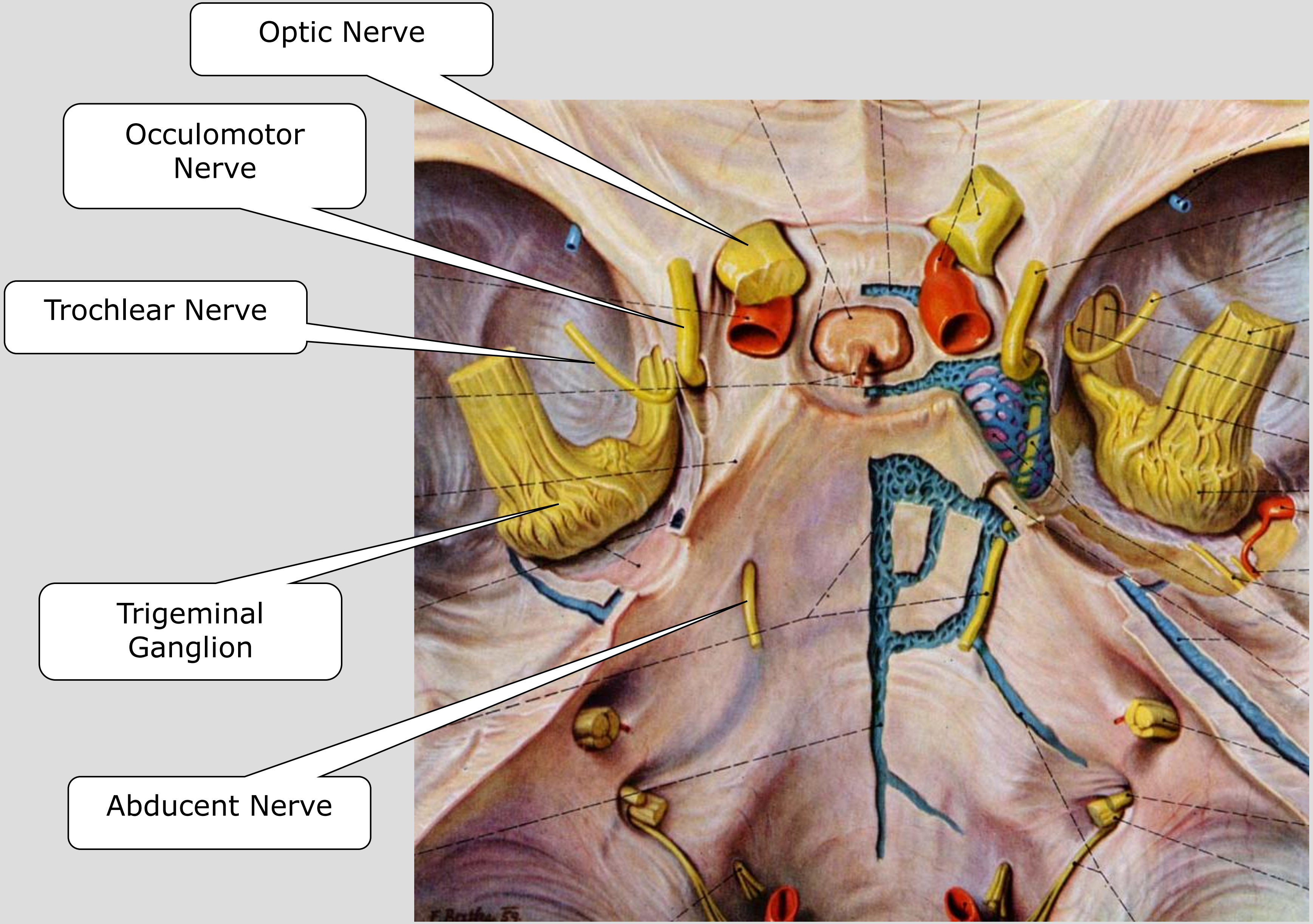
CNV: TRIGEMINAL NERVE

- Path:
  - Coalescence of the four nuclei give rise to the common nerve root that proceeds to the *trigeminal (Gasserian) ganglion, laying in the anterior-lateral aspect of the petrous apex along the temporal bone articulation with the sphenoid.*
  - The ganglion is *incased by thick layers of dura* forming Meckel's cave, from which the three parts of the trigeminal nerve emerge.

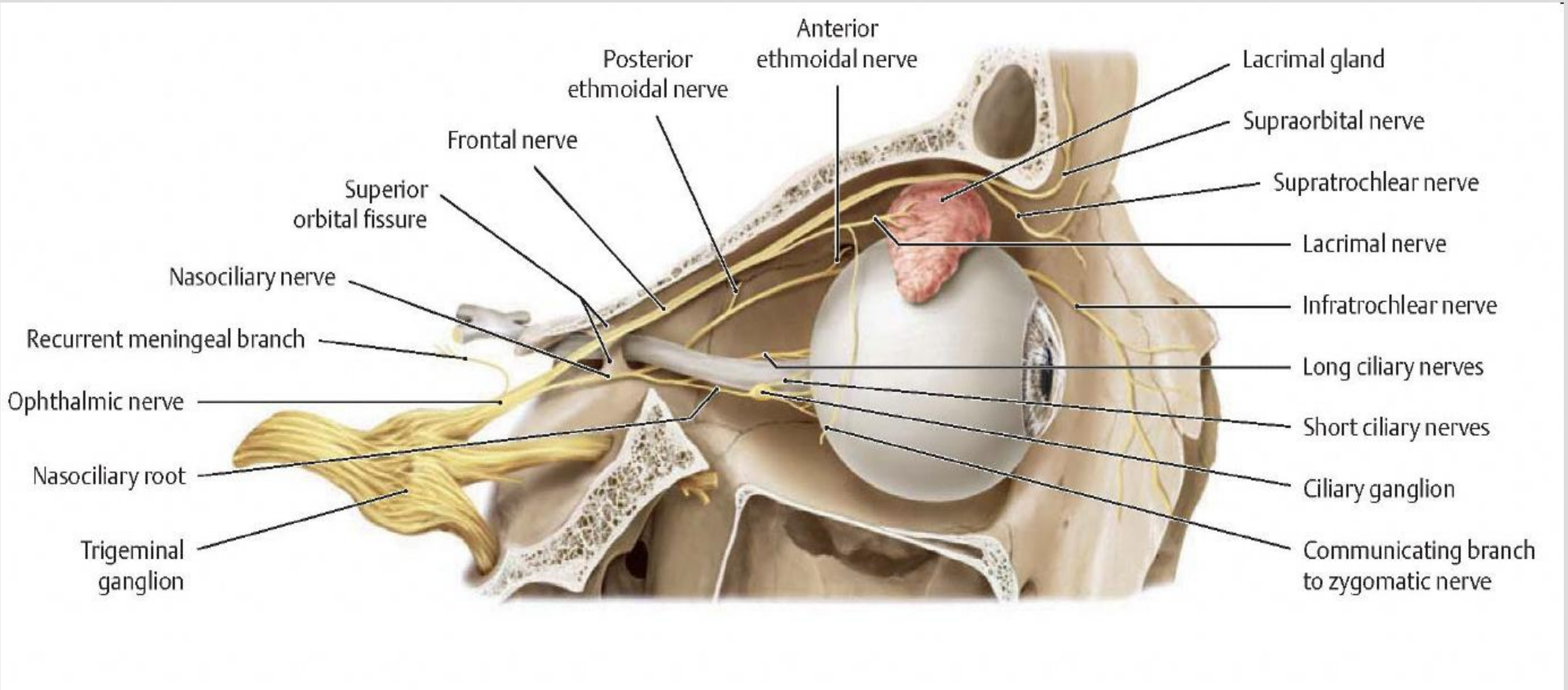
CRANIAL BASE– SUPERIOR VIEW



CNV: TRIGEMINAL NERVE

- Path:
  - *Ophthalmic Division (V_1) –*
    - Proceeds from the *trigeminal ganglion* piercing dura and progresses through the lateral wall of the *cavernous sinus* with branches (recurrent meningeal) to deliver *sensory from the* ***anterior cranial fossa dura mater***.
    - *T*hen through the ***superior orbital fissure*** to divide into the nasociliary, frontal, & lacrimal nerves respectively

OPHTHALMIC DIVISION



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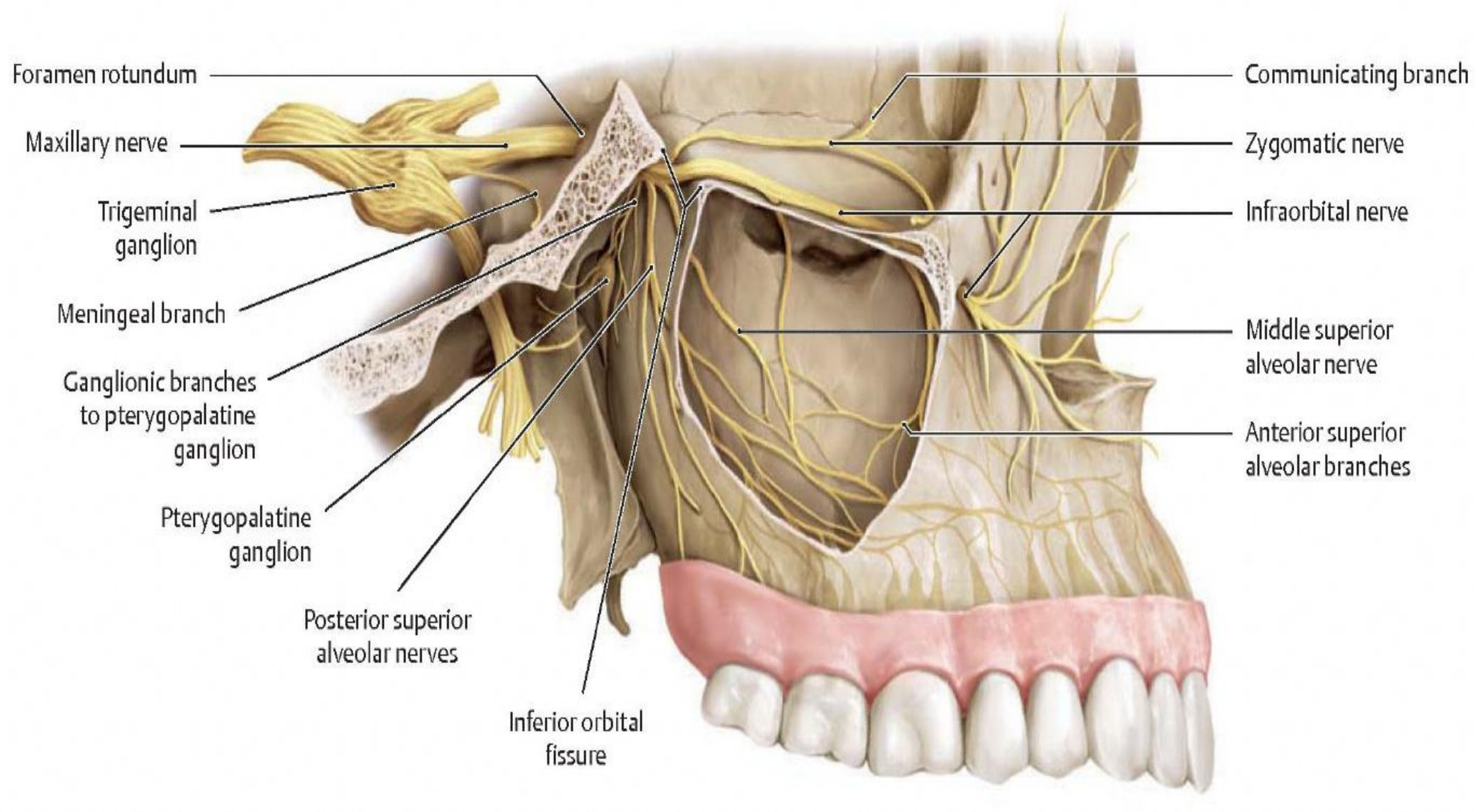
CNV: TRIGEMINAL NERVE

- Path:
  - *Maxillary Division (V_2) –*
    - Proceeds from the trigeminal ganglion over the **sphenosquamous articulation** and it too parts with a meningeal branch delivering sensory from the **middle cranial fossa dura.**
    - Then on through **foramen rotundum** and directly lateral to the orbital process of the palatine bone.
    - Then dividing to form the zygomatic, infraorbital, and alveolar nerves with communications to the **pterygopalatine ganglion** in the pterygopalatine (infratemporal) fossa

MAXILLARY DIVISION



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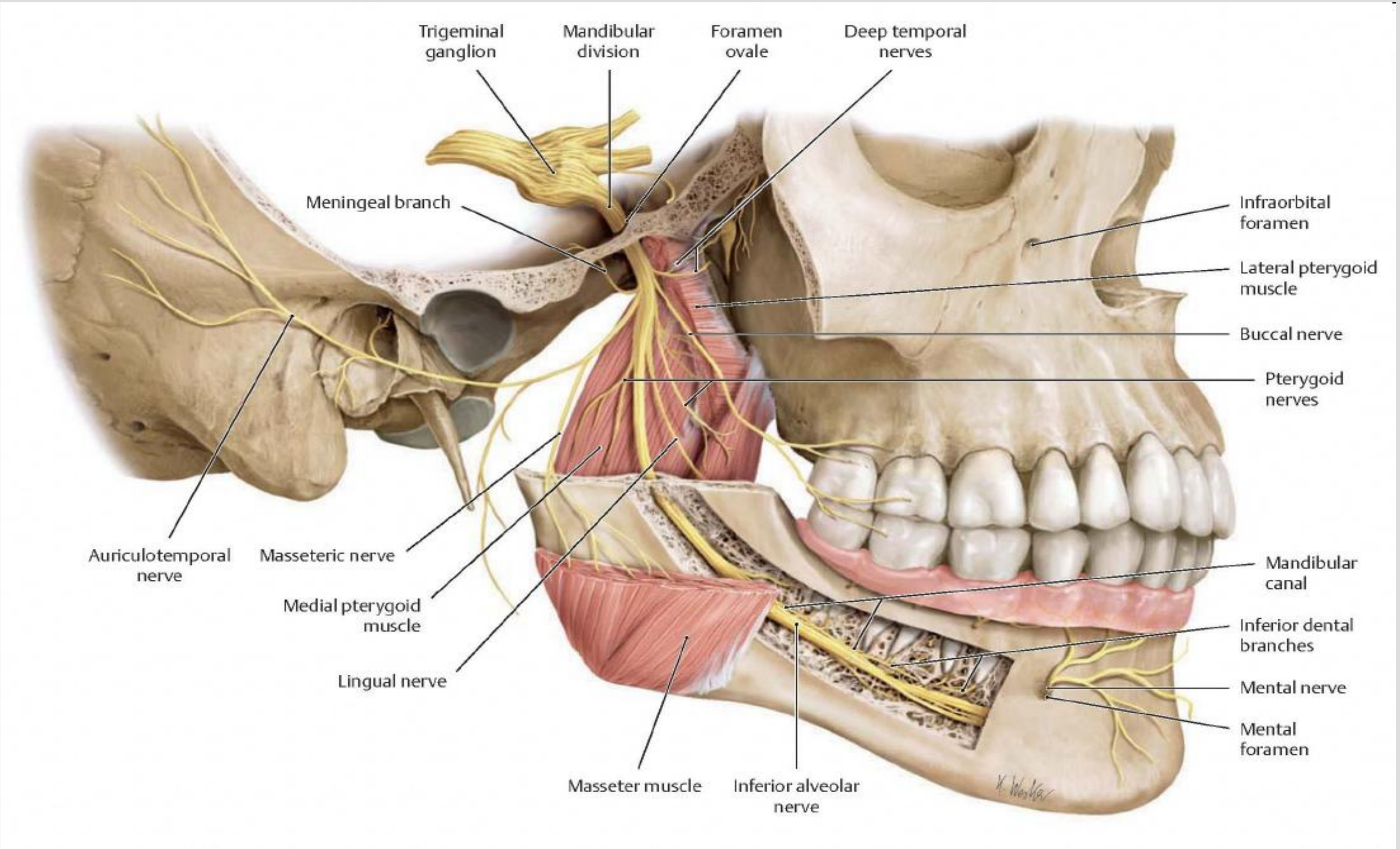
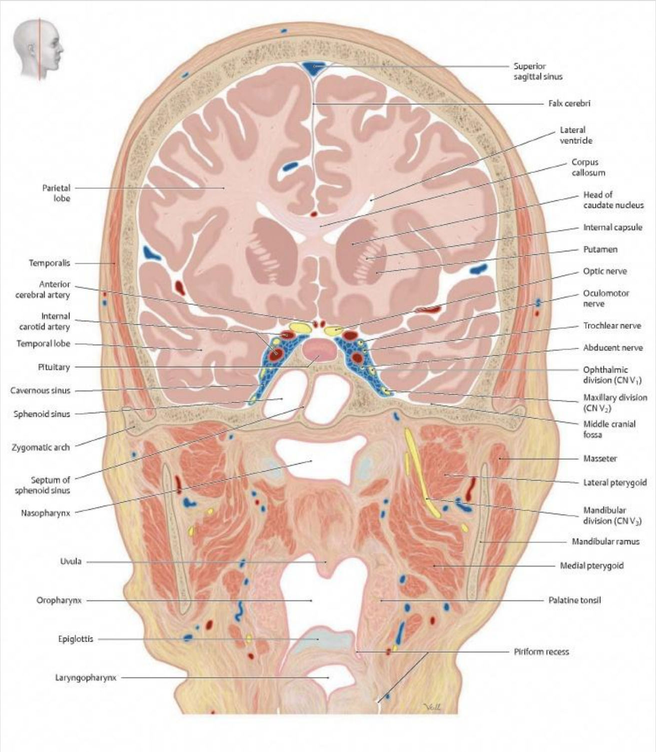
CNV: TRIGEMINAL NERVE

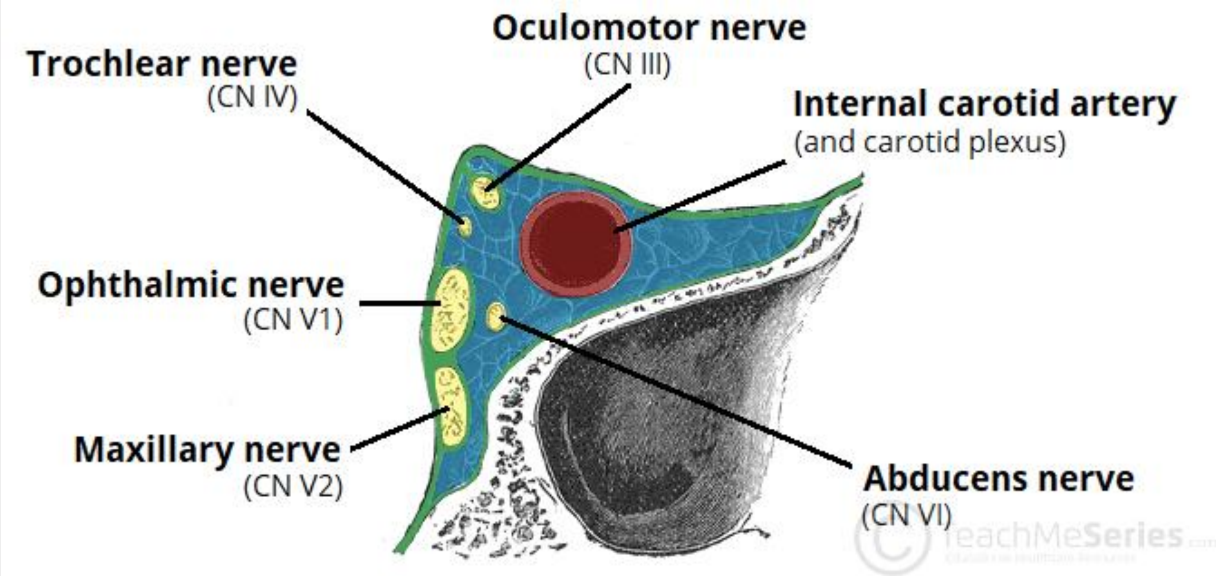
- Path:
  - Mandibular Division (V_3) –
    - Proceeds over the **sphenosquamous articulation** then through **foramen ovale**.
    - **After which it too gives off a sensory meningeal branch** that renters through foramen spinosum to innervate the **middle cranial fossa dura.**
    - V_3 then proceeds into the infratemporal fossa splaying forth its **sensory and motor** branches to the dentition, mucosa, auditory areas and skin as well as motor for muscles of mastication

MANDIBULAR DIVISION



CAVERNOUS SINUS





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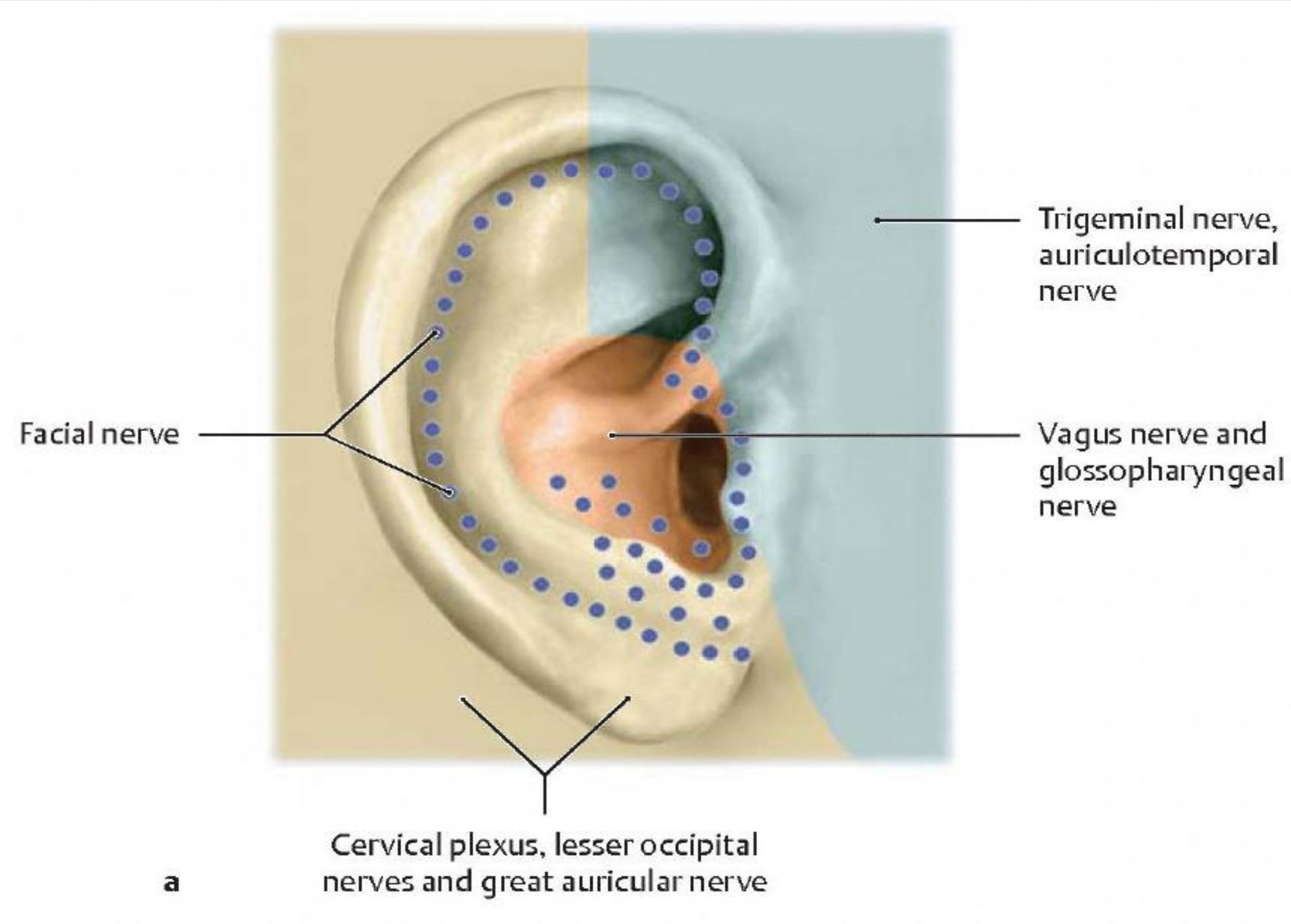
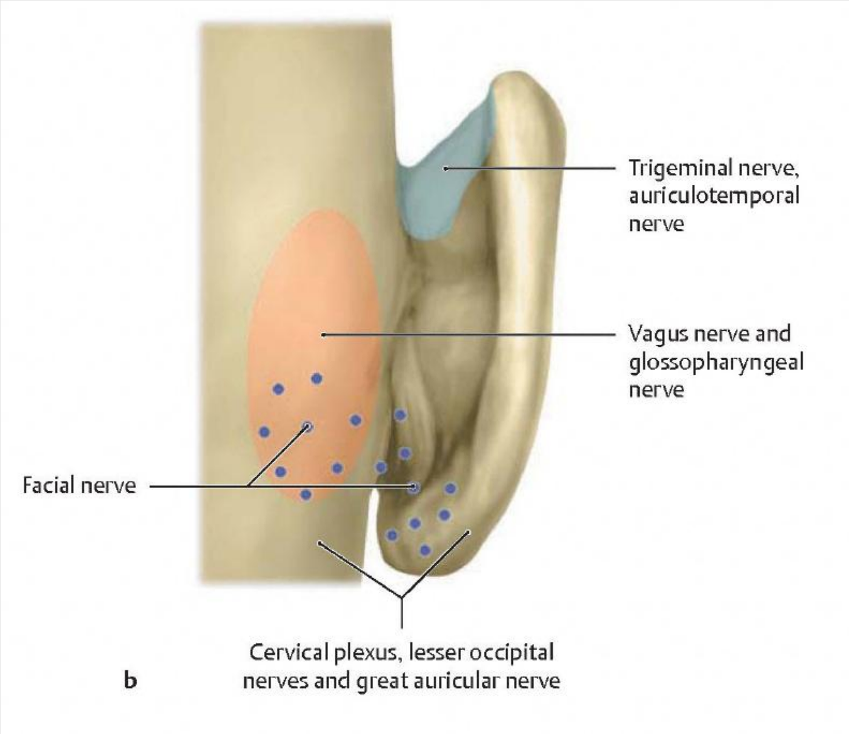
PUTTING THE EAR DERMATOME TOGETHER





CN V: TRIGEMINAL NERVE

- Clinical/Osteopathic Considerations:
  - Meckel's cave is susceptible to **dural strains** (frequently traumatic) focally between the **petrosphenoidal articulations** or more distally through dural interconnectivity. Symptoms may manifest as **facial pain, and/or cephalgia**
  - V_2 & V_3 with their path over the anterior **sphenosquamous articulations** is considered a possible point of irritation to those roots
  - V_1 ,V_2,V_3 sensory respectively enters the cranium at the supraorbital fissure/foramen, infraorbital fissure, and mental foramen respectively, thus allowing for a **palpatory inflammatory assessment** of their related tissue (Sinuses etc …)
  - Dural irritations of anterior and middle cranial fossa are considered closely related to V_1,V_2,V_3, and middle meningeal artery inflammation –possible etiology of **migraine headaches**

CN VII: FACIAL NERVE

- Origin:
  - Superior salivatory nucleus
  - Facial Nucleus
  - Nucleus of solitary tract
- Fiber Types:
  - Visceral afferent:
    - Gustatory form the anterior 2/3 of tongue
  - Special visceral efferent:
    - Muscle of facial expression and stapedius
    - Parasympathetic to glandular tissue

CN VII: FACIAL NERVE

- Path:
  - Emerges from the cerebellopontine region entering the **internal auditory (acoustic) meatus**, giving off the **Greater Petrosal** nerve at the geniculate ganglion to exit the temporal bone at the **Hiatus Fallopii**
  - Then joining the **Deep Petrosal nerve** with connections from the internal carotid artery and plexus.
  - After the **Greater Petrosal nerve** exits the Petrosal canal (Videan canal within the sphenoid) it synapses on the pterygopalatine ganglion (Infratemporal Fossa) .
  - These fibers then proceed to the lacrimal gland and the nasal glands through a **shared pathway with V_2**.
  - The Facial nerve continues to gives off the **Stapedial nerve** and then the **Chorda Tympani** nerve before exiting from the **Stylomastoid Foramen**.
  - After exiting, it innervates the **stylohyoid** and **posterior belly of the digastric muscles** as well as dividing into the respective branches to innervate the **facial muscles**

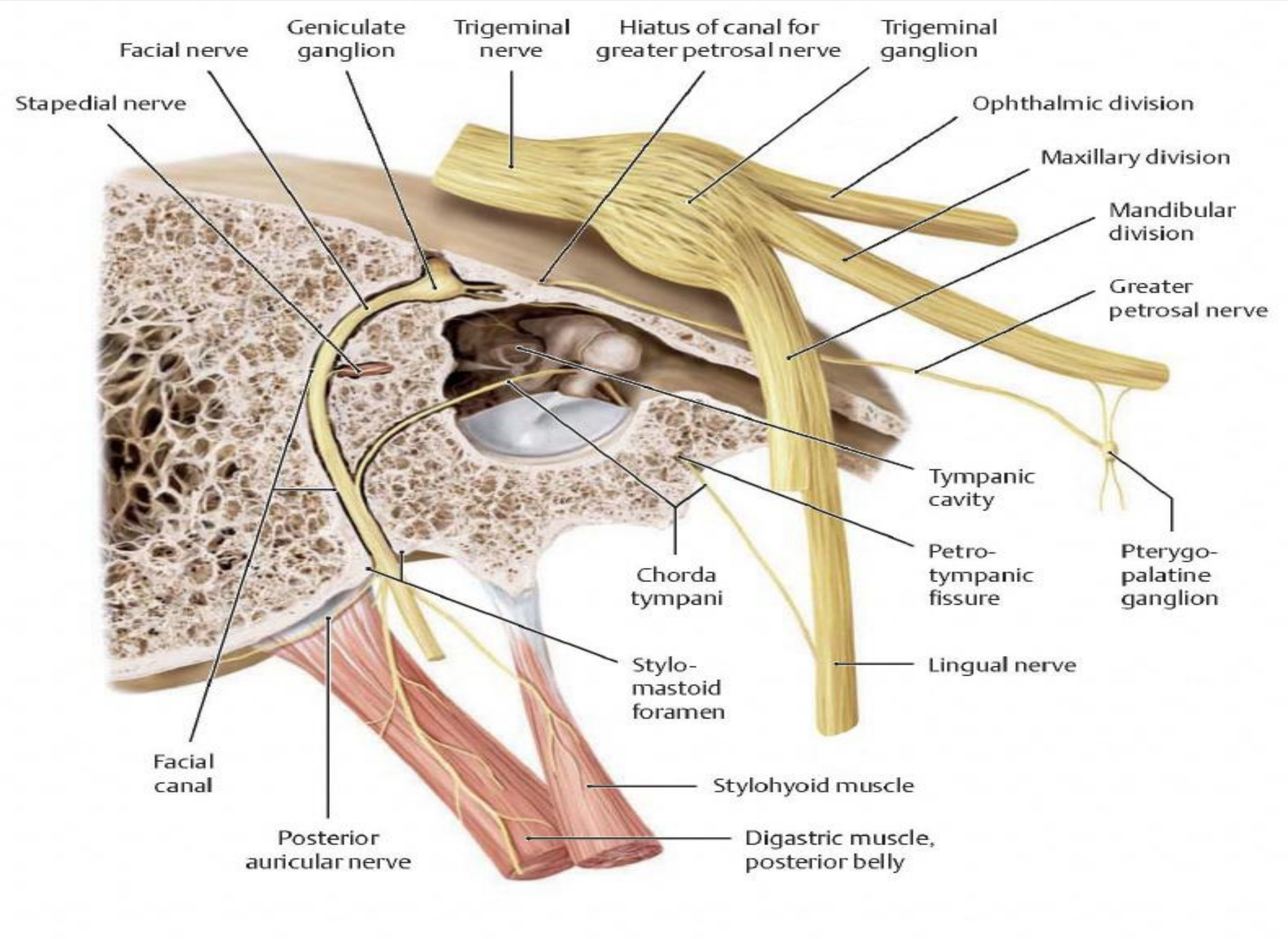
CN VII: FACIAL NERVE

- Path Continued:
  - The **Chorda Tympani** exits the temporal bone through the **Petrotympanic Fissure** with fibers synapsing in the **submandibular ganglion** to innervate the **submandibular** , **sublingual**, and **other small glands** of the palate, mucosa and dorsum of the tongue.
  - **Afferent gustatory (taste) fibers** from the tongue separate as the Chorda Tympani nerve from the shared neural pathway of the **V_3 lingual nerve branch**, back through the Petrotympanic Fissure to the middle ear and joining with CN VII. From there it synapses on the **solitary tract nucleus.**

CNVII: FACIAL NERVE

- Clinical/Osteopathic Considerations:
  - The nerve's intimate association with the **temporal bone** causes close scrutiny of that bone's functional status within the primary respiratory mechanism
  - Dysfunction can lead to poor host responses to inflammatory etiologies of **facial nerve neuropathies**
  - Symptoms: hearing, taste, salivary and lacrimal gland secretions, as well as facial muscle dysfunction
  - Additionally, temporal mandibular joint (TMJ) dysfunction can be associated with the above symptoms due its close proximity of the chorda tympani exit at the petrotympanic fissure

FACIAL NERVE CONSIDERATIONS WITHIN THE PETROUS PORTION OF THE TEMPORAL BONE



Illustrator: Karl Wesker pp. 78-79

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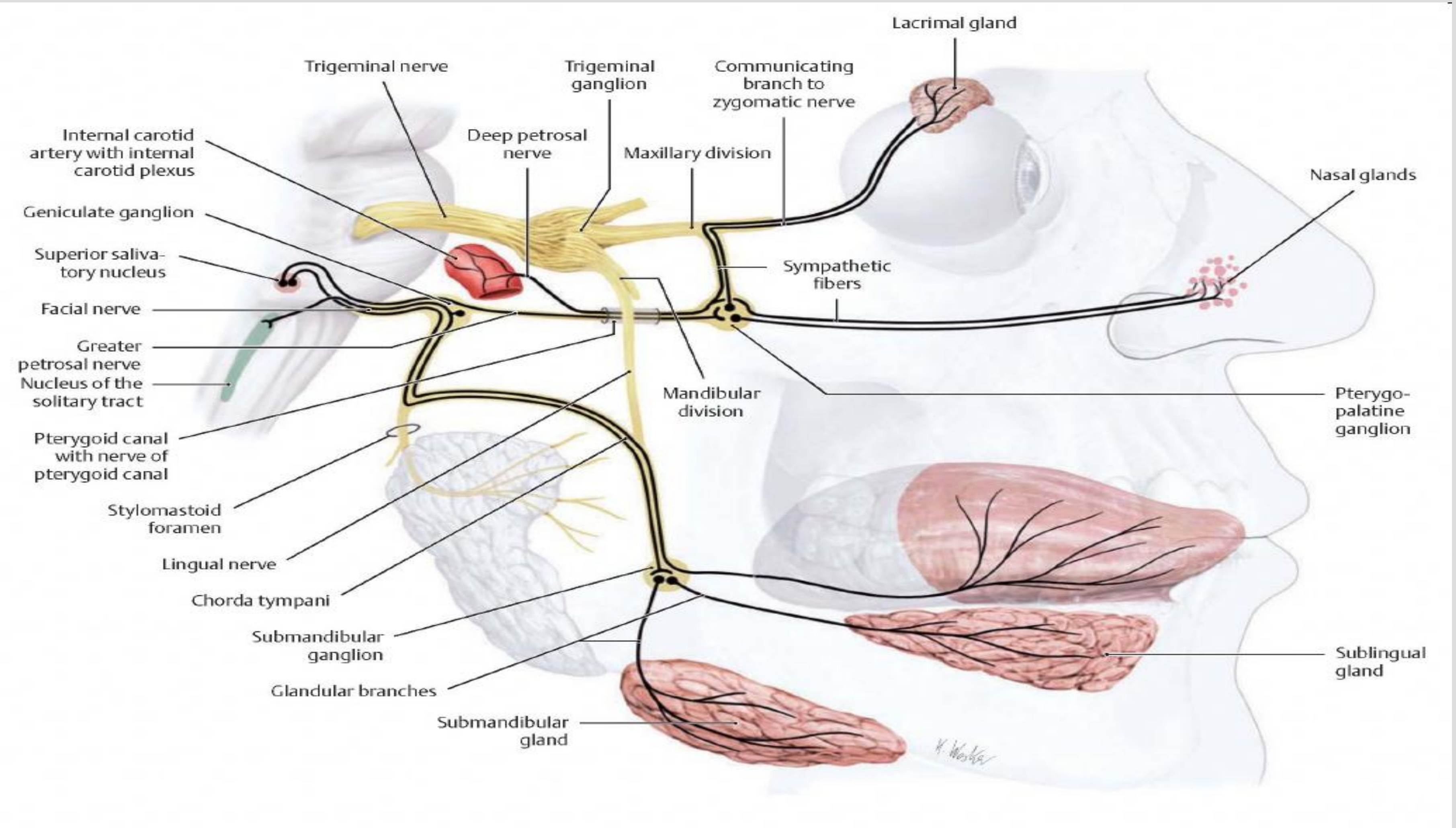
PARASYMPATHETIC & SYMPATHETIC EFFERENTS OF THE FACIAL NERVE



C Parasympathetic visceral efferents and visceral afferents (gustatory fibers) of the facial nerve

Illustrator: Karl Wesker pp. 78-79
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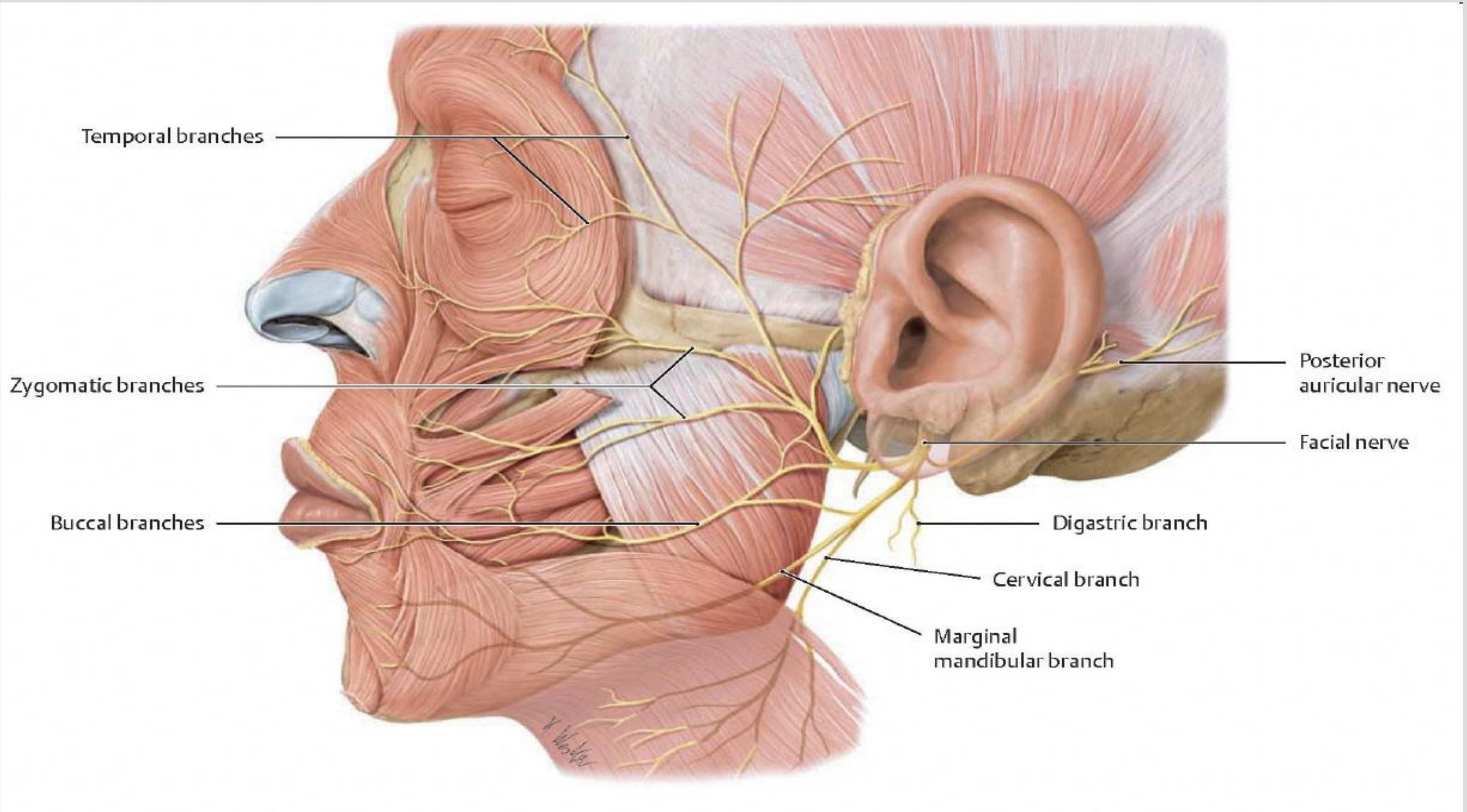
FACIAL MUSCLE INNERVATIONS



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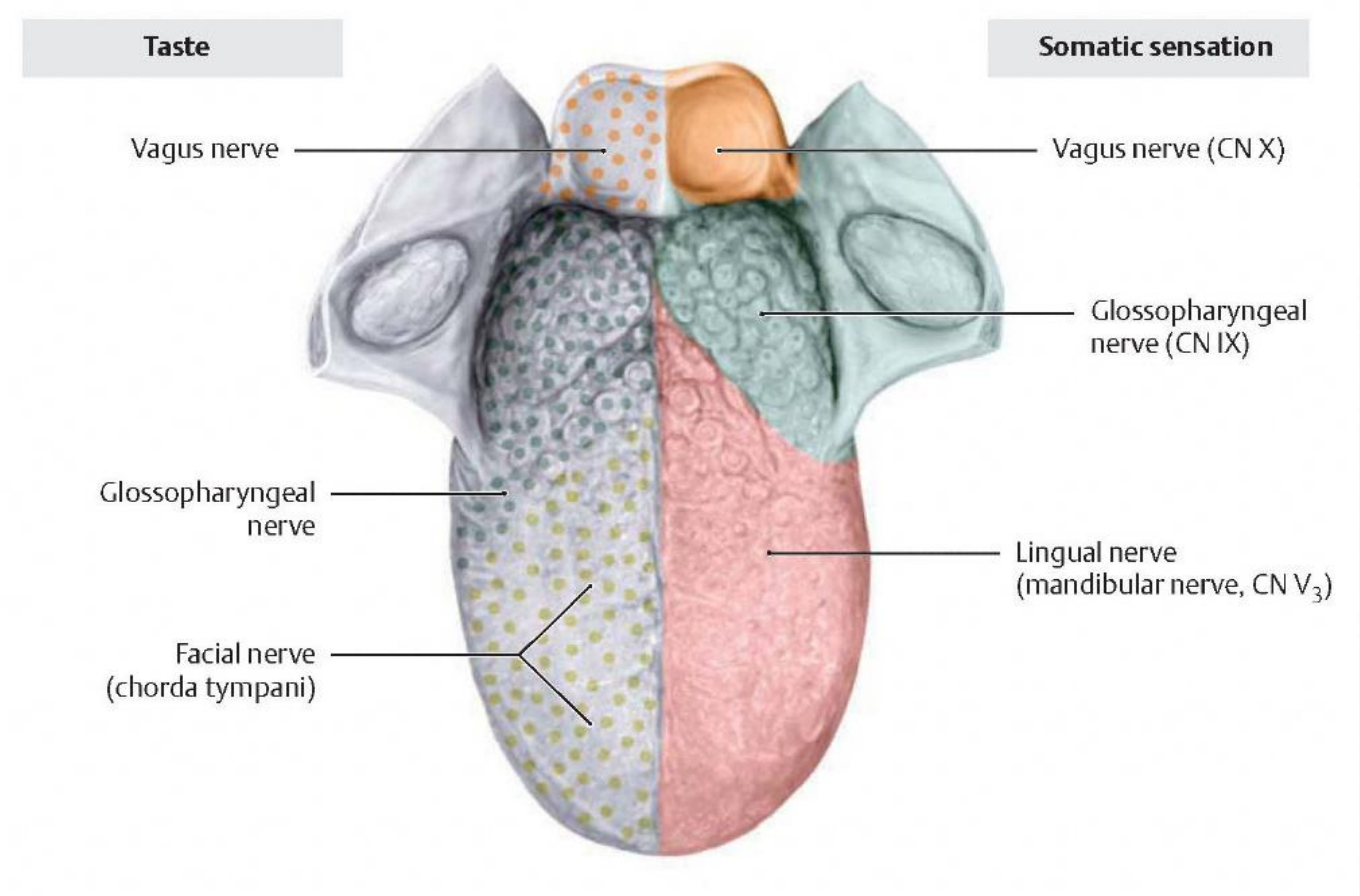
PUTTING THE TONGUE TOGETHER



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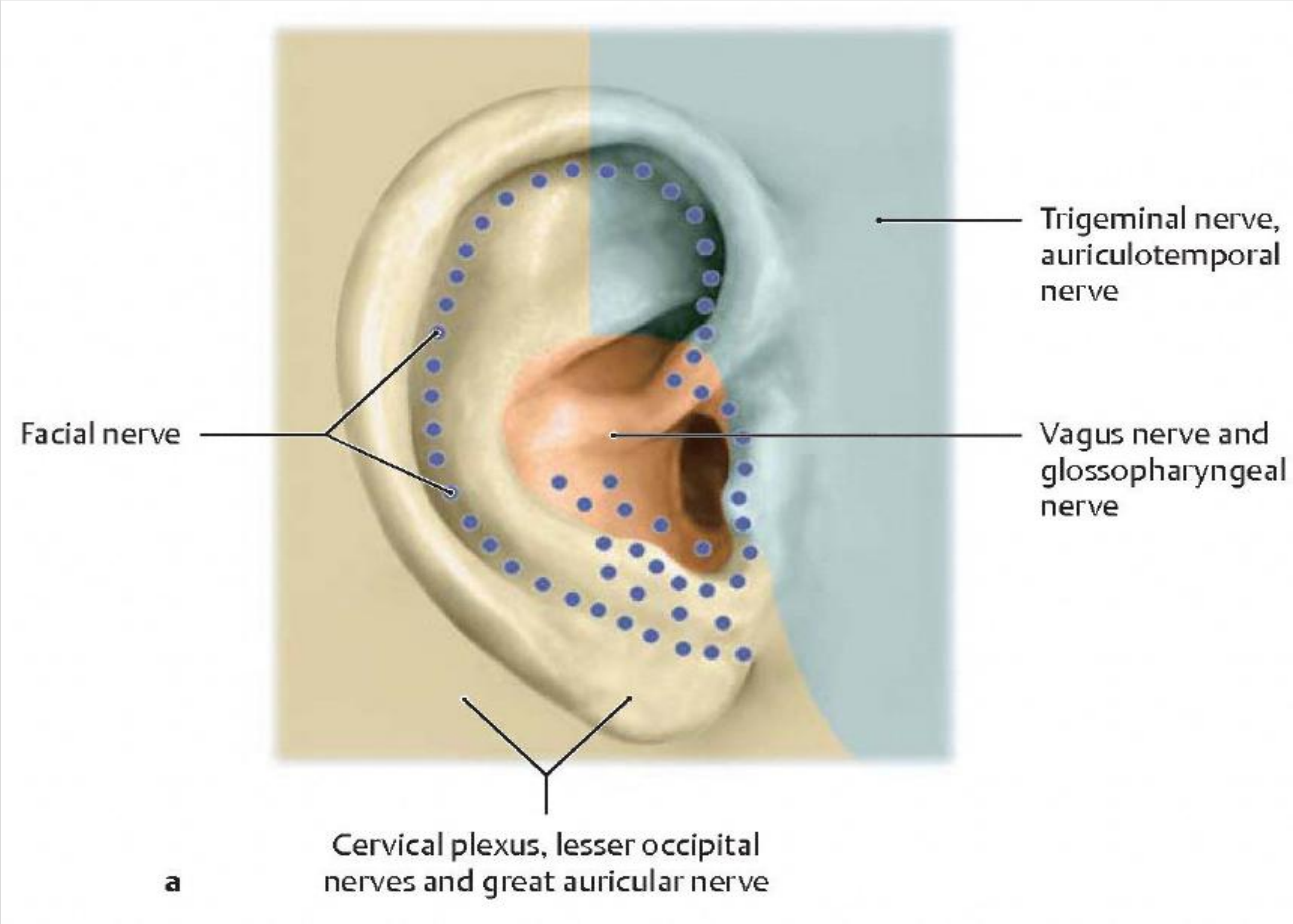
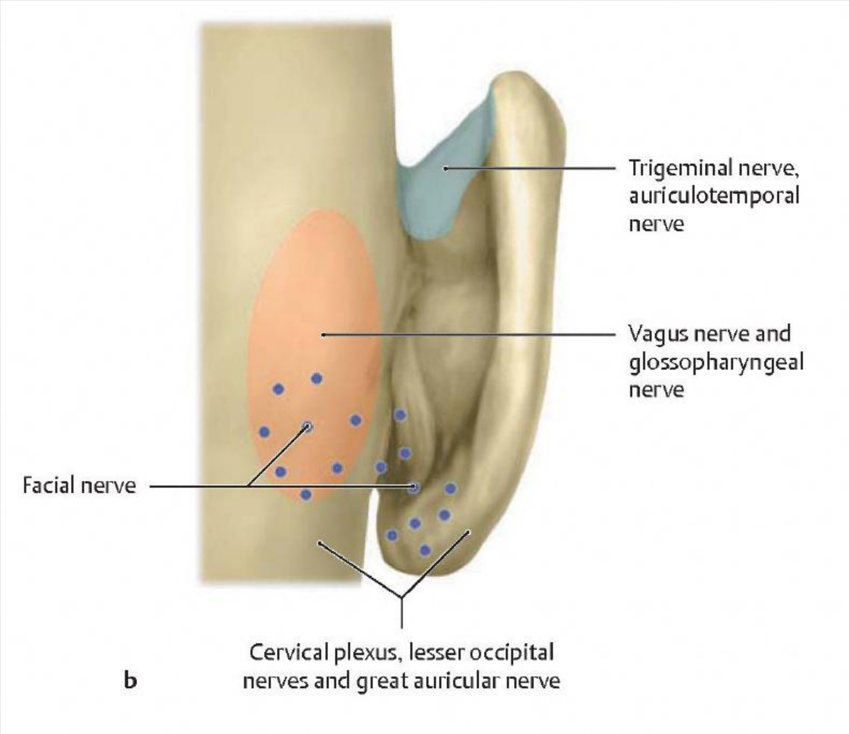
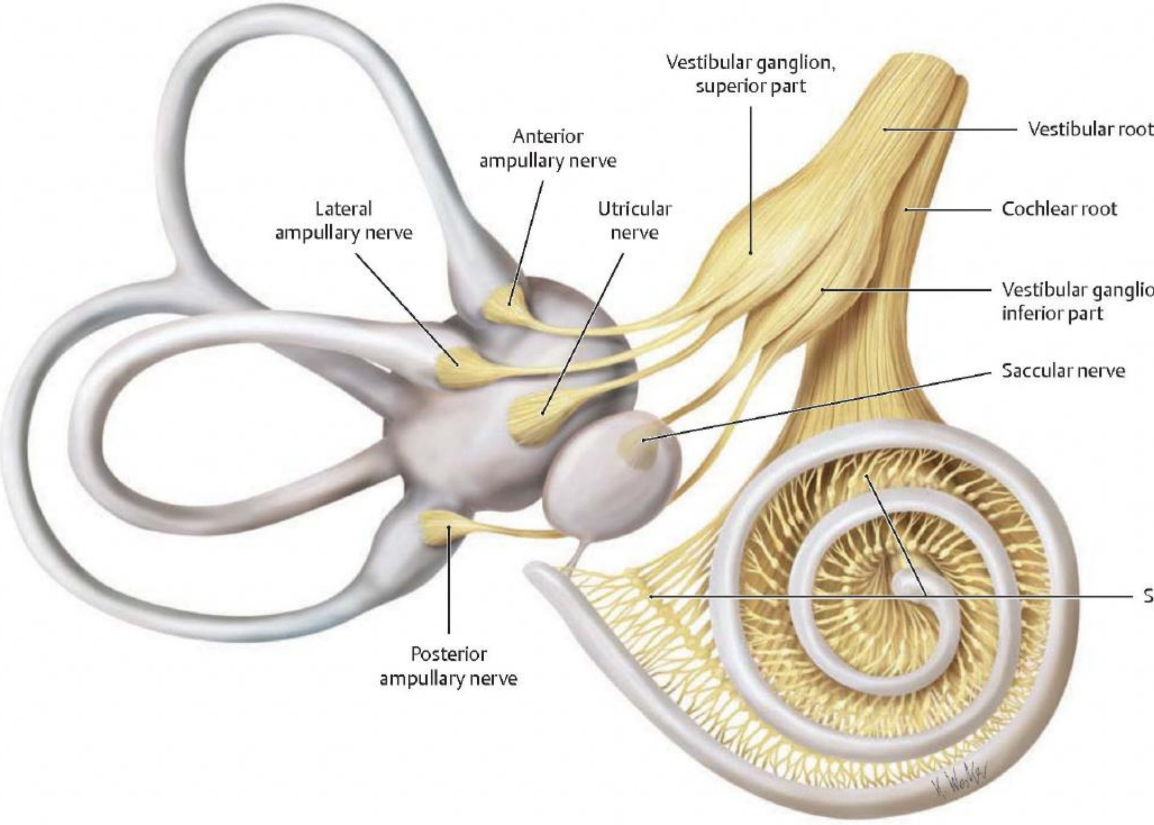
PUTTING THE EAR DERMATOME TOGETHER

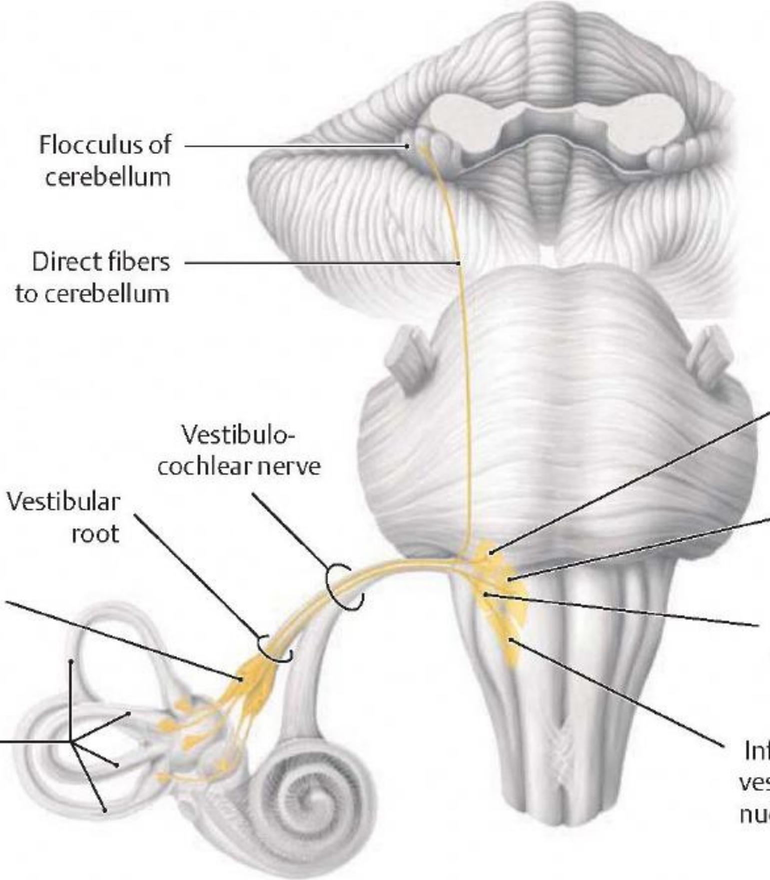




CN VIII: VESTIBULOCOCHLEAR NERVE

- Origin:
  - Vestibular Nucleus (4)
  - Cochlear Nucleus (2)
- Fiber Types:
  - Special Somatic Afferent:
    - Vestibular: gravitational equilibrium
    - Cochlear: auditory
- Path:
  - Following their afferent pathway, *they share a common dural sheath* as they pass out the internal auditory meatus to their respective nuclei at the cerebellopontine angle
- Clinical/Osteopathic Considerations:
  - *Synchronous physiologic temporal bone motion is critical for proper equilibrium information* as well as *proper pressure gradients of the inner, middle, and external ear. Dural strains can be present at the IAM to alter neural axonal health*
  - *Tinnitus and vertigo are common symptoms*





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Illustrator: Karl Wesker
pp. 82-83
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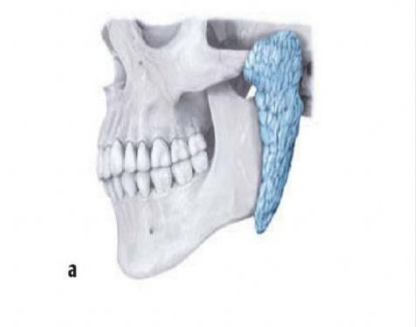
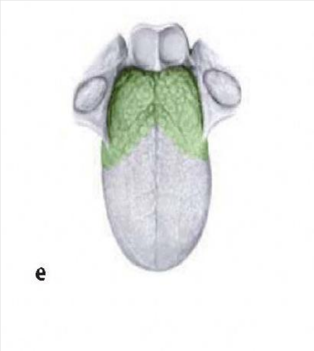
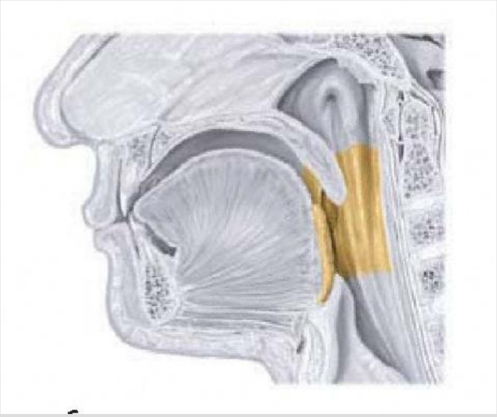
CN IX: GLOSSOPHARYNGEAL NERVE

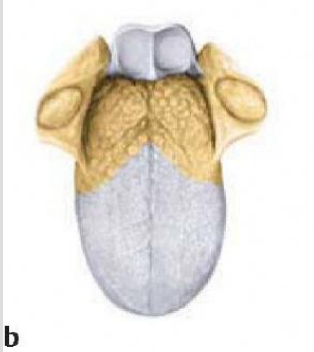
- Origin:
  - Inferior salivatory nucleus
  - Nucleus ambiguous – shared with CN X & XI
  - Solitary tract nucleus
  - Trigeminal spinal nucleus – shared with CN V & X
- Fiber Types:
  - Special visceral efferent
    - Joins with pharyngeal plexus for pharyngeal constrictor muscles
  - General visceral efferent
    - *Parasympathetic to the parotid gland , uses branches of V_3 (auriculotemporal nerve)*
  - Somatic afferent
    - *Sensory from the post 1/3 of tongue, soft palate, & pharyngeal mucosa (gag reflex)*
    - Tympanic cavity and Eustachian tube mucosa
    - External ear and external auditory canal sensory (shared with CN X)
  - Special visceral afferent
    - *Gustatory from the posterior 1/3*
  - General visceral afferent
    - *Chemo and baroreceptors from the carotid body*

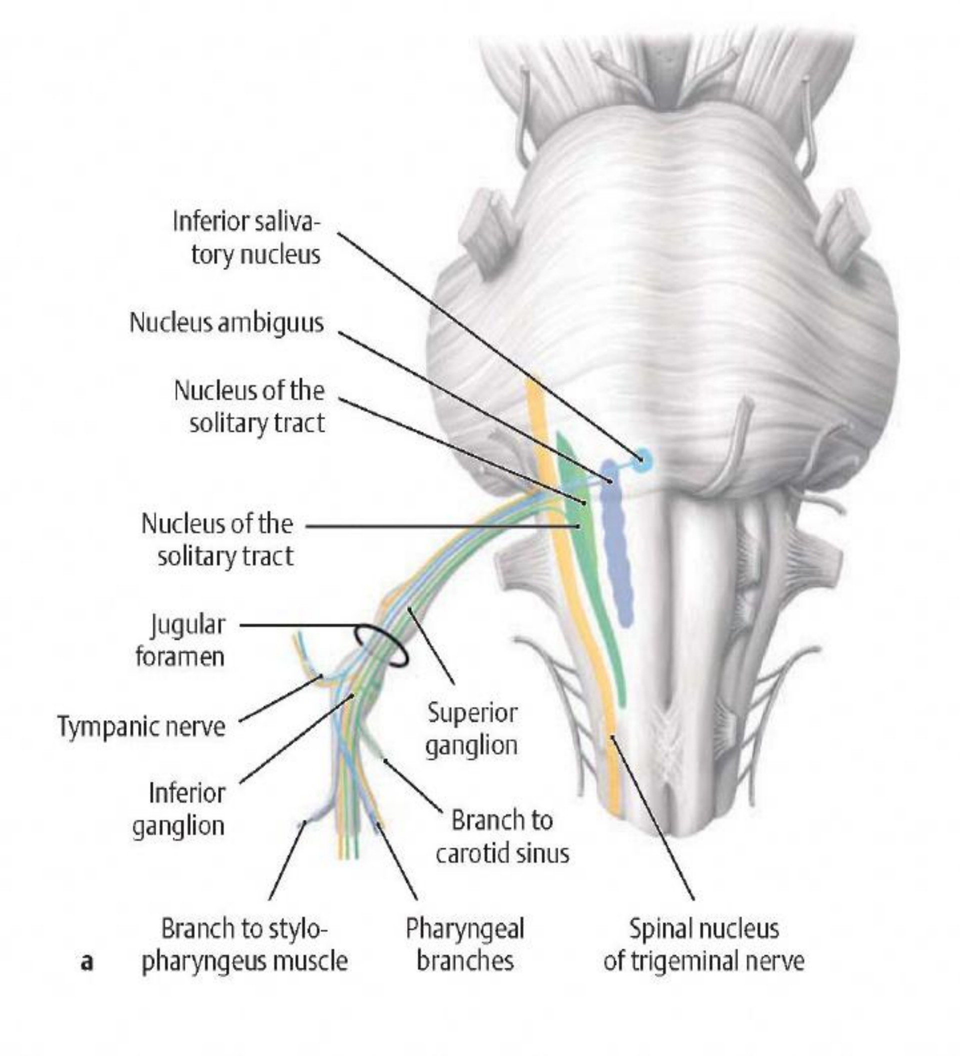
GLOSSOPHARYNGEAL NERVE







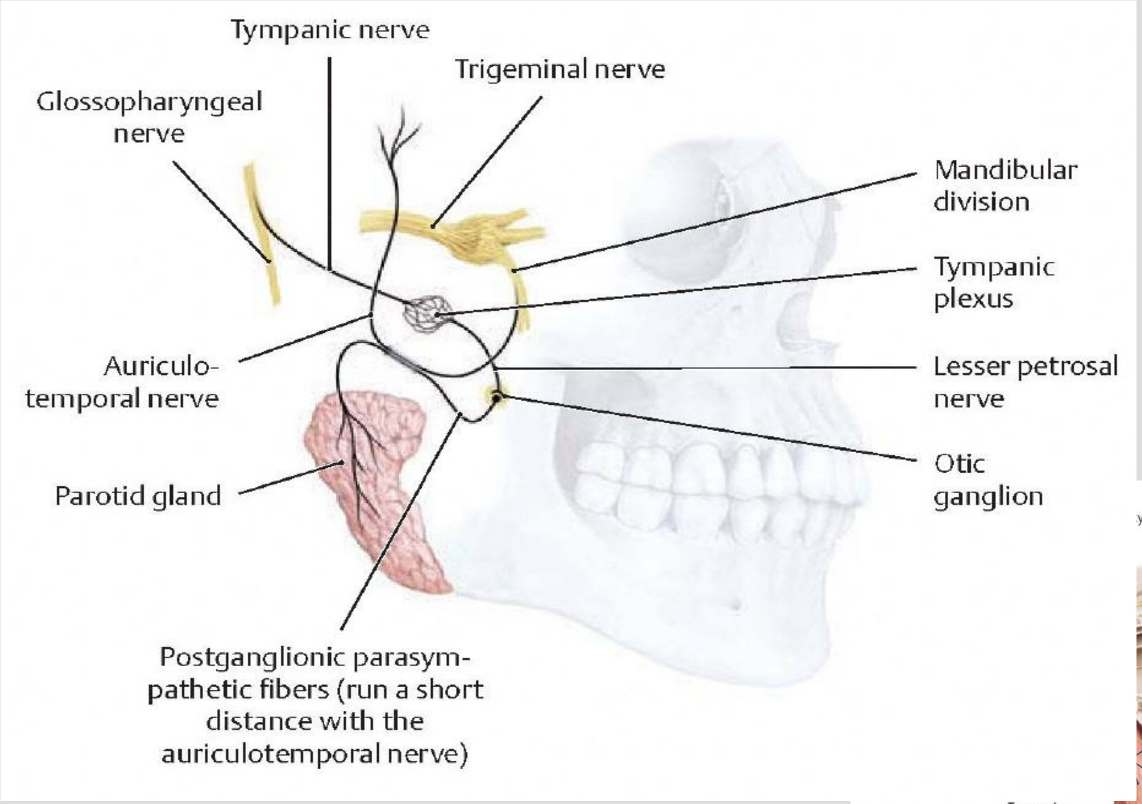


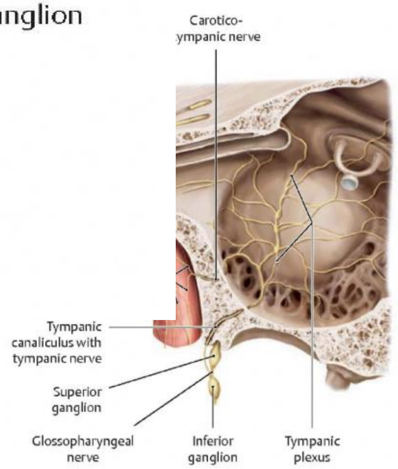


CN IX: GLOSSOPHARYNGEAL NERVE

- *Path*
  - Emerges from the medulla oblongata to *exit the cranium via the jugular foramen.*
  - Note: it is associated with two glossopharyngeal ganglions - the superior and inferior that are separated by the jugular foramen
- *Clinical/Osteopathic Considerations*
  - Neuropathy symptoms are characterized by swallowing difficulties whether in newborn, child or adult. Consider central nervous system verse peripheral in dysfunction. With peripheral, issues at the Jugular Foramen, Occipitomastoid Suture or Occipital Condyles should be considered.

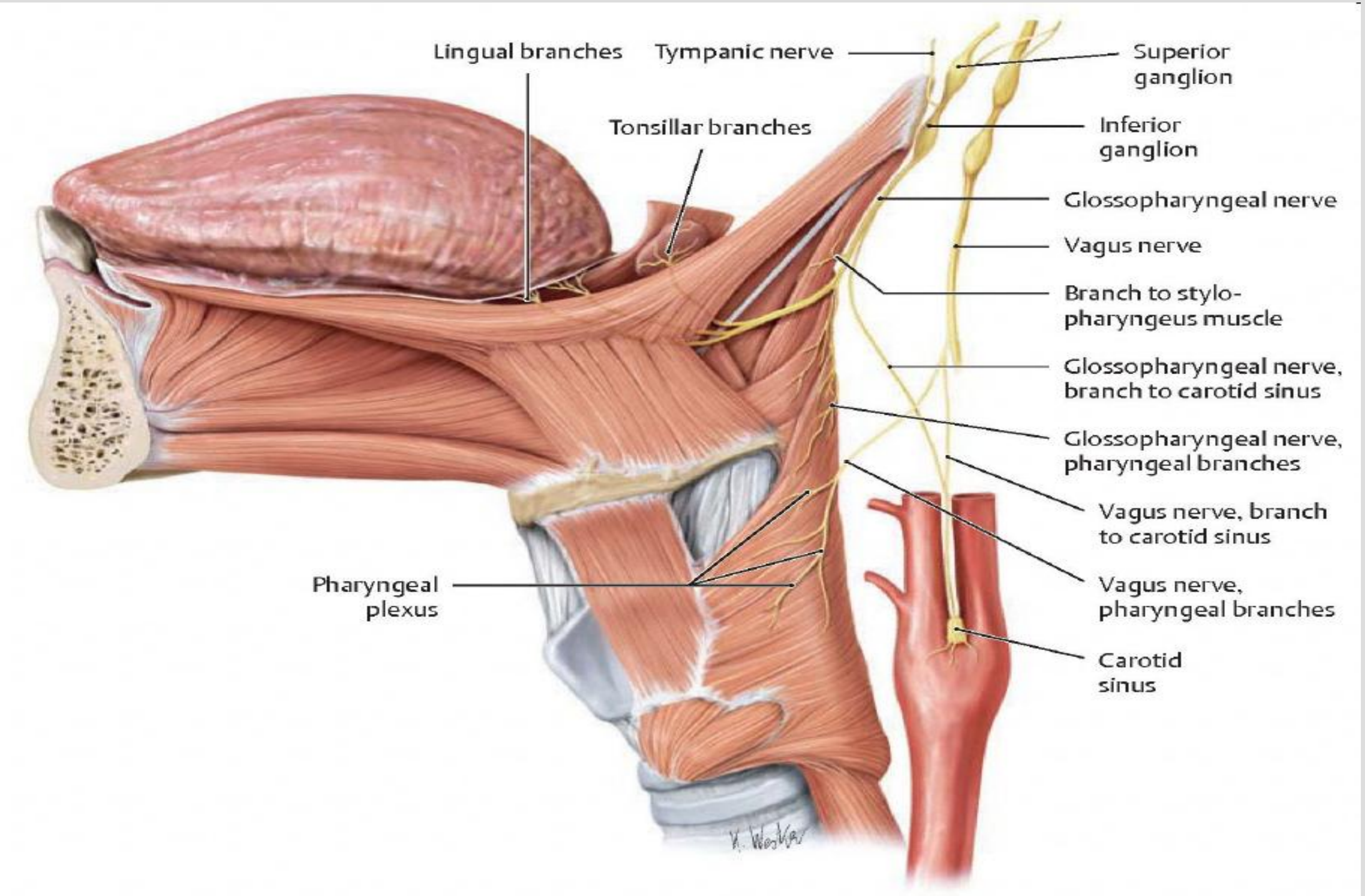
GLOSSOPHARYNGEAL NERVE





D Branches of the glossopharyngeal nerve in the tympanic cavity
Left anterolateral view.

GLOSSOPHARYNGEAL NERVE



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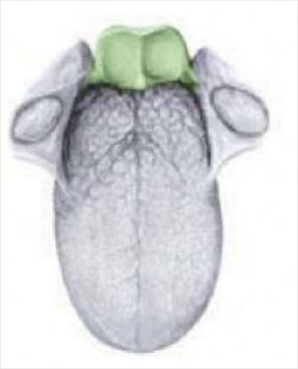
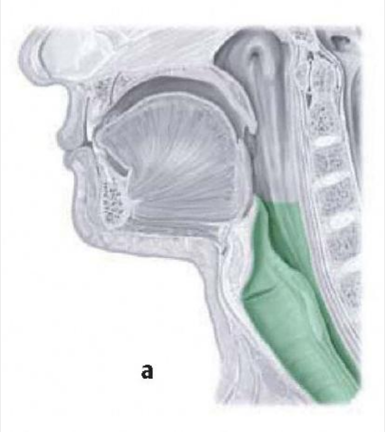
CNX: VAGUS NERVE

- Origin:
  - Dorsal vagal nucleus
  - Nucleus ambiguous – shares portions with CNs IX & XI
  - Nucleus of the solitary tract – shares portions with CNs VII & IX
  - Trigeminal spinal nucleus – shares portions with CNs V & IX
- Fiber Types
  - Special visceral efferent
    - *Joins the pharyngeal plexus to innervate the pharyngeal constrictors in collaboration with CN IX*
    - Laryngeal muscles
  - General visceral efferent
    - *Parasympathetic to the thoracic and abdominal viscera to as far as the mid-transverse colon*
  - Somatic afferent
    - Sensory from the *dura of the posterior cranial fossa*
    - *Sensory from the auricular region* as well as the *external auditory canal (shared with CN IX)*
  - Special visceral afferent
    - Gustatory buds on the epiglottis (who knew?)
  - General visceral afferent
    - Sensory from the *lower pharynx & esophagus*
    - Laryngeal mucosa above and below the glottic aperture
    - *Chemo and baroreceptors from the aortic arch and para-aortic body (see CN IX)*
    - *Sensory from the thoracic and abdominal viscera*

VAGUS NERVE

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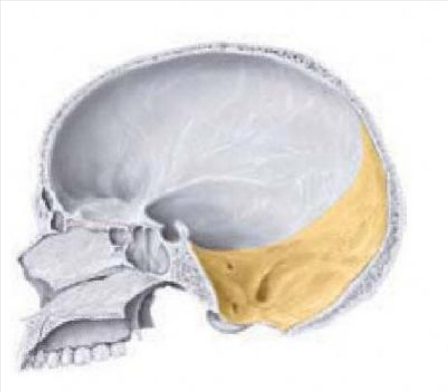
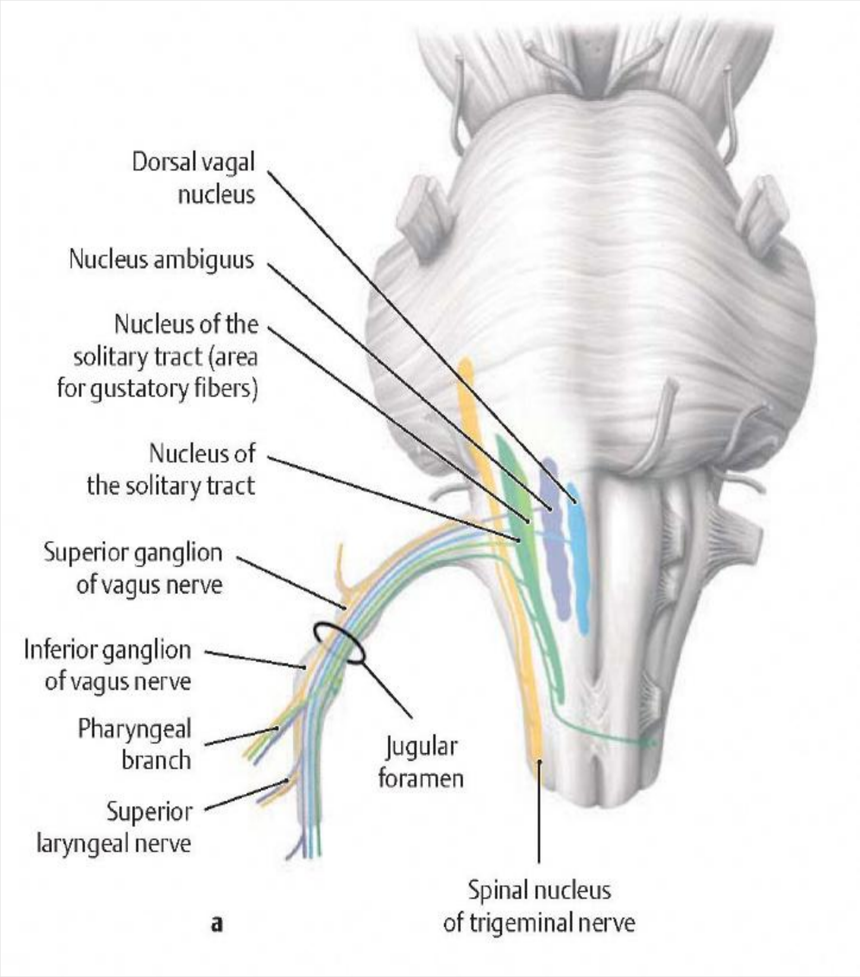
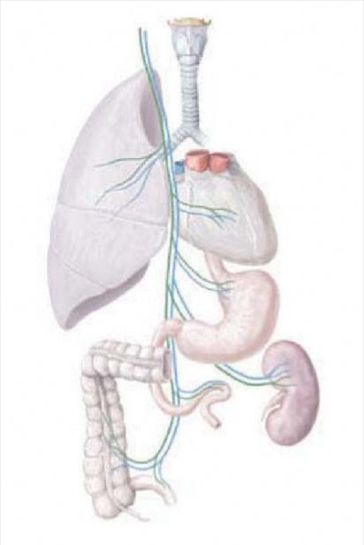
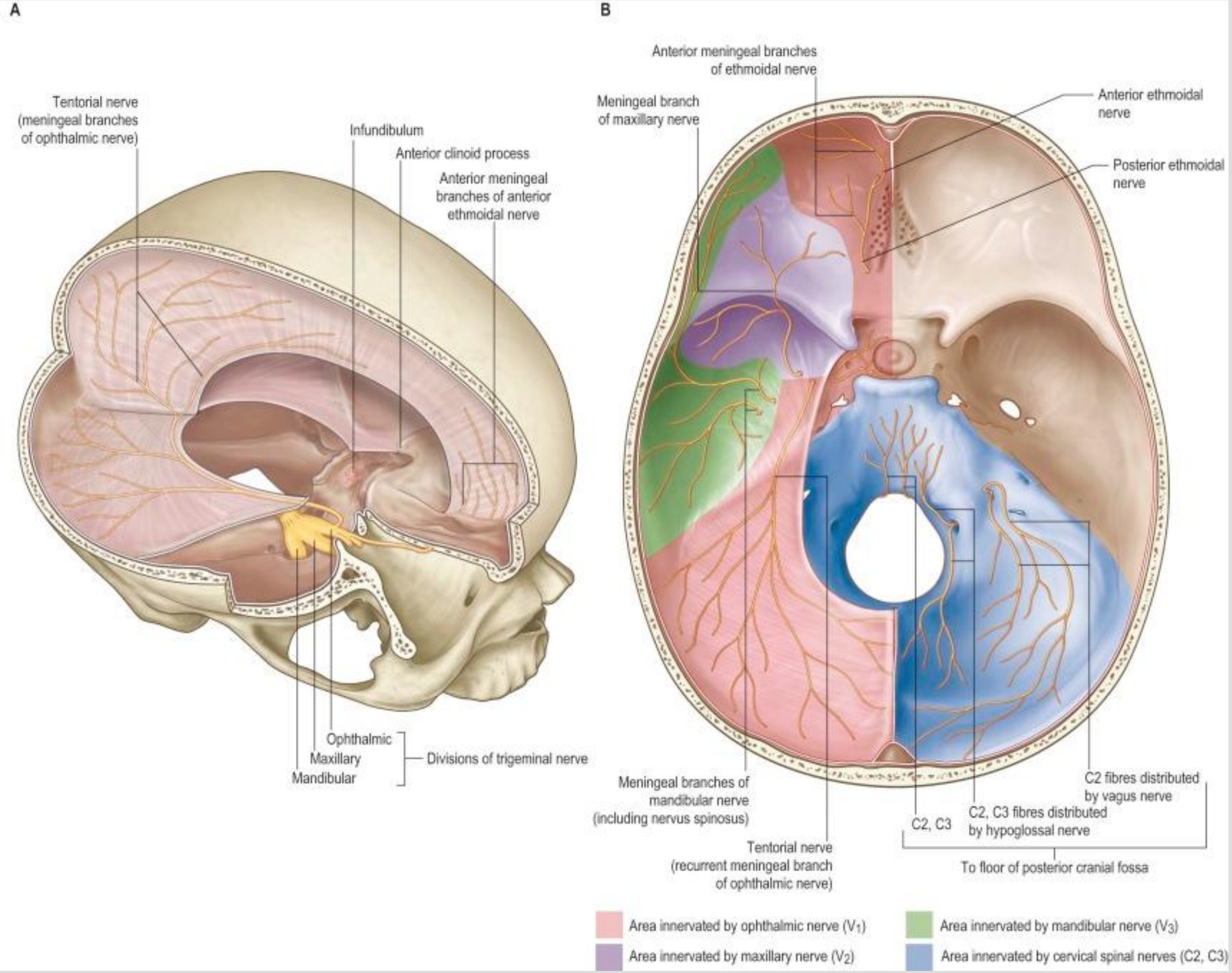



Dorsal vagal nucleus
Nucleus ambiguus
Nucleus of the solitary tract (area for gustatory fibers)
Nucleus of the solitary tract
Superior ganglion of vagus nerve
Inferior ganglion of vagus nerve
Pharyngeal branch
Superior laryngeal nerve
Jugular foramen
Spinal nucleus of trigeminal nerve
a

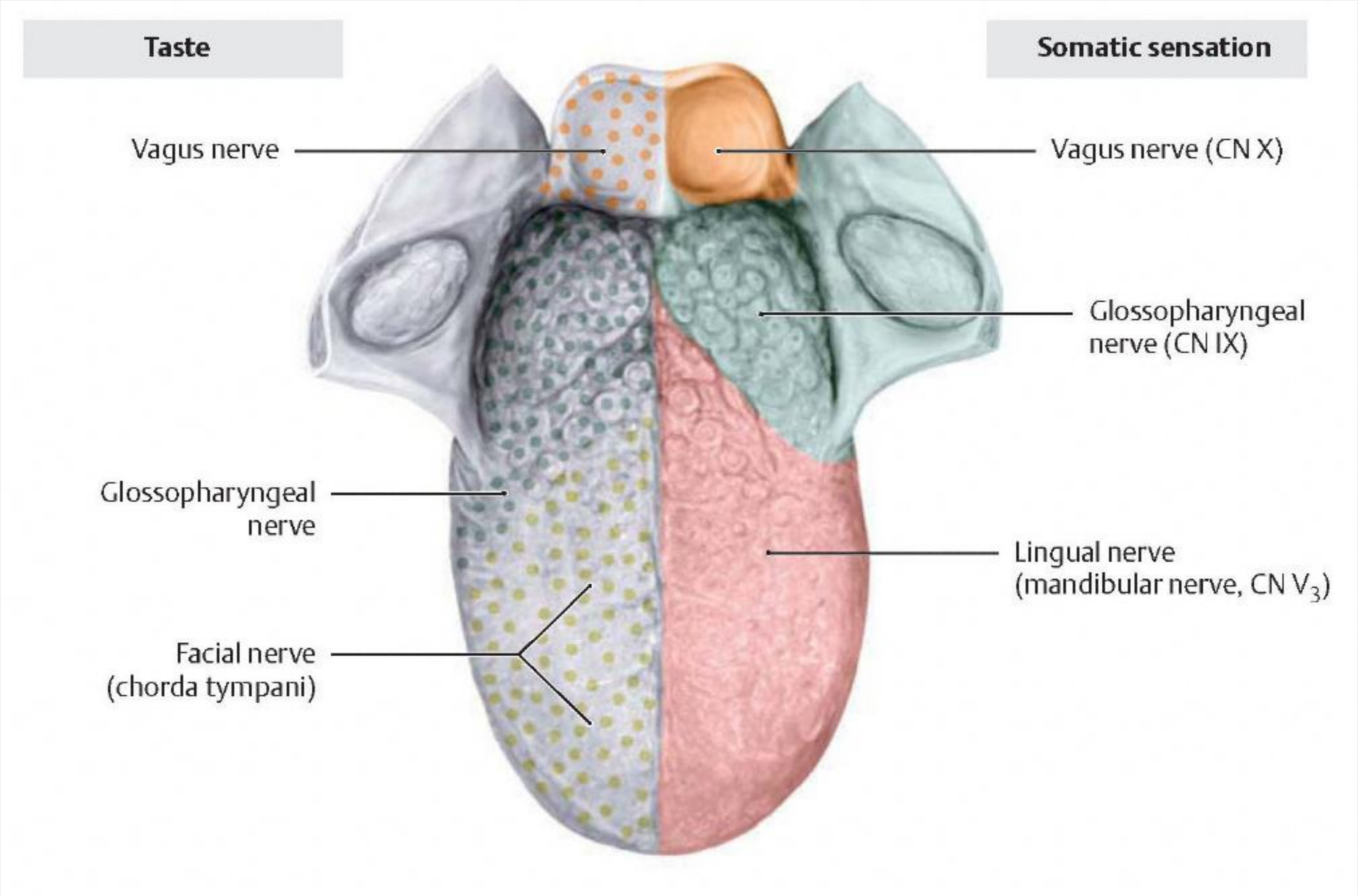



A
Tentorial nerve
(meningeal branches
of ophthalmic nerve)
Infundibulum
Anterior clinoid process
Anterior meningeal
branches of anterior
ethmoidal nerve
Ophthalmic
Maxillary
Mandibular
Divisions of trigeminal nerve
B
Anterior meningeal branches
of ethmoidal nerve
Meningeal branch
of maxillary nerve
Anterior ethmoidal
nerve
Posterior ethmoidal
nerve
Meningeal branches of
mandibular nerve
(including nervus spinosus)
Tentorial nerve
(recurrent meningeal branch
of ophthalmic nerve)
C2, C3
C2, C3 fibres distributed
by hypoglossal nerve
C2 fibres distributed
by vagus nerve
To floor of posterior cranial fossa
Area innervated by ophthalmic nerve (V_1)
Area innervated by maxillary nerve (V_2)
Area innervated by mandibular nerve (V_3)
Area innervated by cervical spinal nerves (C2, C3)

CNX: VAGUS NERVE

- *Path (Vagus as in Vagabond)*
  - Emerges from the medulla oblongata *dispersing meningeal branches* before *exiting the cranium through the jugular foramen* after which it enters the carotid sheath as it proceeds on its wandering path.
  - Note that here too (as in CN IX) it has two ganglion that are separated by the jugular foramen
- *Clinical Osteopathic Considerations*
  - Issues concerning this nerve are considerable (as you might imagine) *Newborn sucking dysfunction & post prandial emesis are seen, as well as Cardiac dysrythmia, esophageal spasm, and respiratory irregularities.*
  - Follow the path and consider the organ parasympathetic innervation. Again, consider central nervous system verse peripheral in dysfunction. With peripheral, *issues at the jugular foramen, occipitomastoid suture or occipital condyles must be considered.*
  - It behooves one to remember that *proper physiology is achieved through a balance of the sympathetic with the parasympathetic neural activity*

PUTTING THE TONGUE TOGETHER



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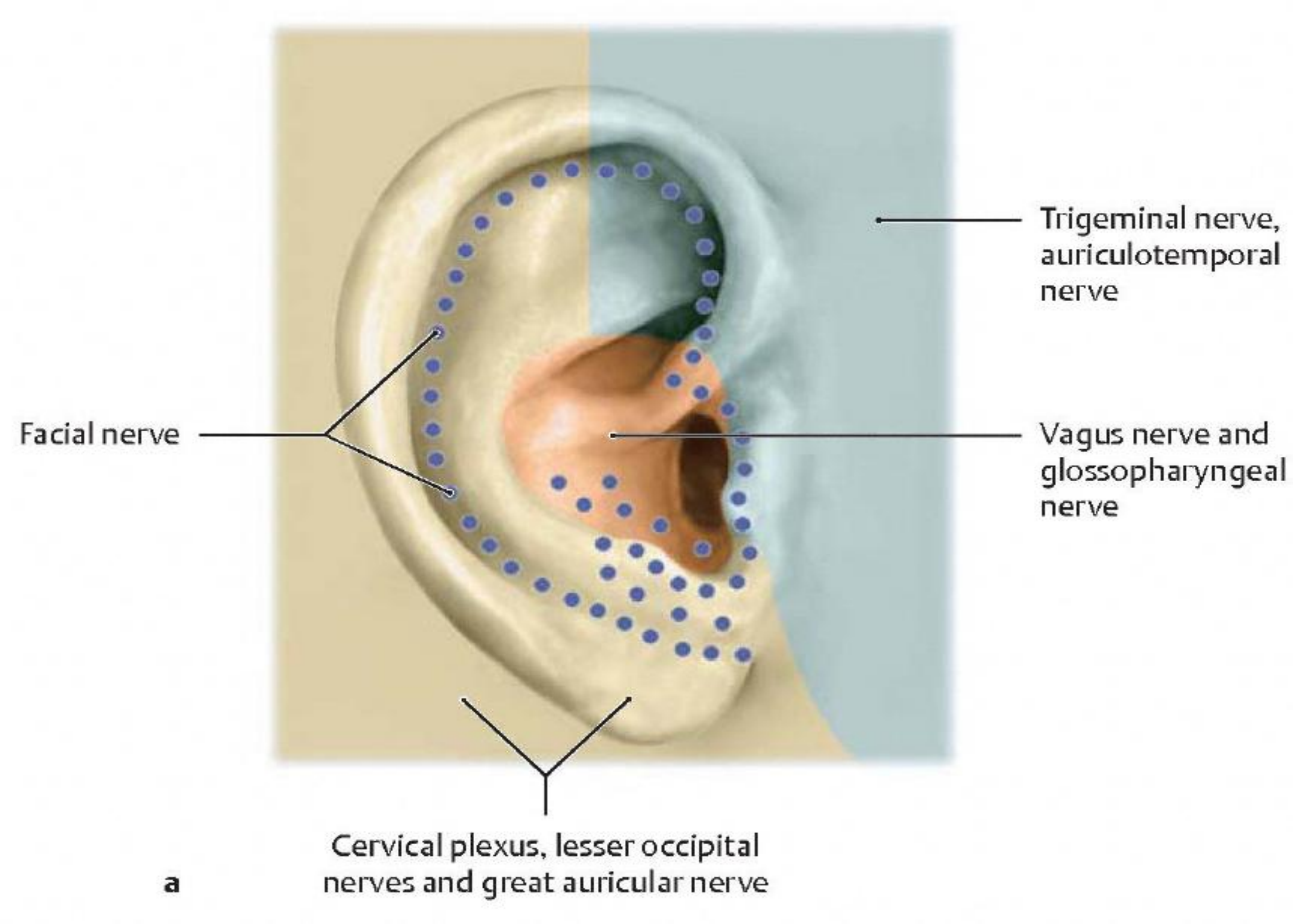
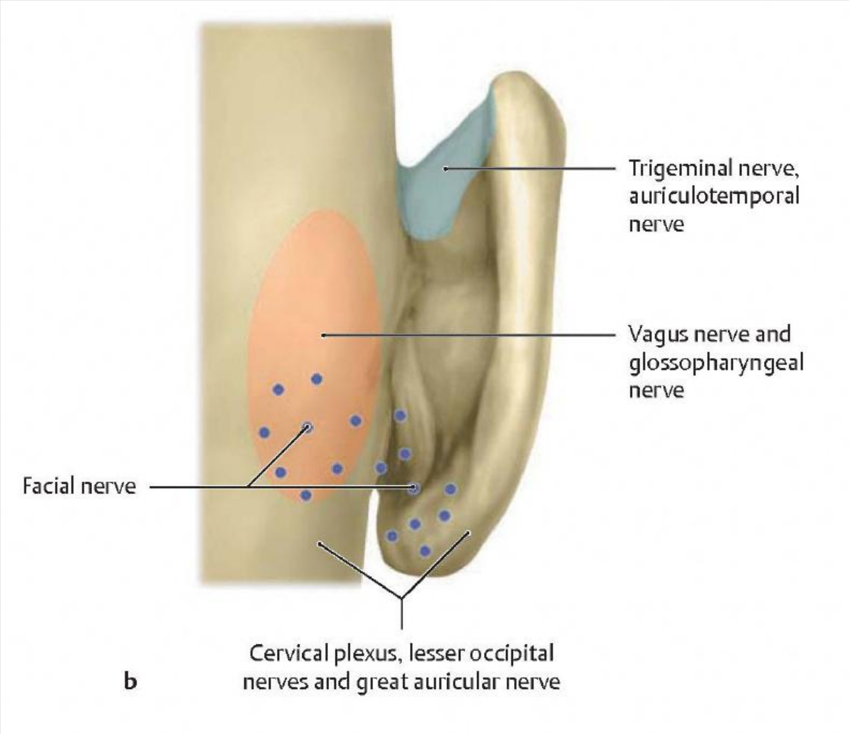
PUTTING THE EAR DERMATOME TOGETHER

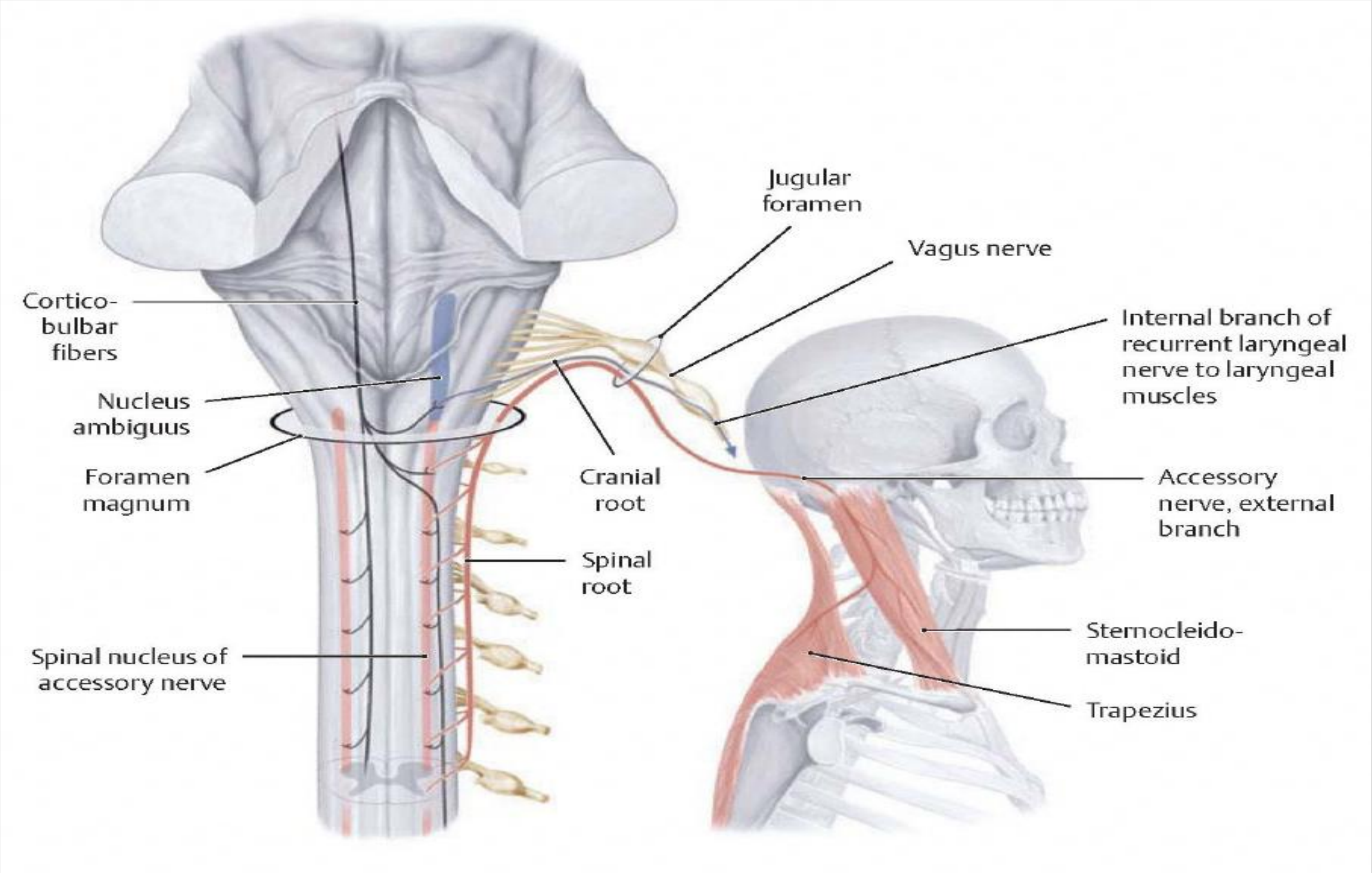




CN XI: SPINAL ACCESSORY NERVE

- Origin:
  - Spinal nucleus of the accessory nerve
  - Nucleus ambiguous – shared portions with CN IX & X
- Fiber Types
  - Special visceral efferent
    - Laryngeal muscles except cricothyroid muscles
  - General somatic efferent
    - Trapezius & sternocleidomastoid muscles
- Path
  - Cranial root proceeds from the nucleus ambiguous and joins the vagus nerve and then descends through the jugular foramen to innervate the laryngeal muscles by way of the recurrent laryngeal nerve
  - Spinal roots proceed as a collective of nerves emanating from the anterior horn of levels C2 – C5/6 ascending up through foramen magnum joining CNs IX & X to exit the cranium through the jugular foramen to synapse on the Trapezius and Sternocleidomastoid muscles.

(SPINAL) ACCESSORY NERVE (CN XI)



CN XI: SPINAL ACCESSORY NERVE

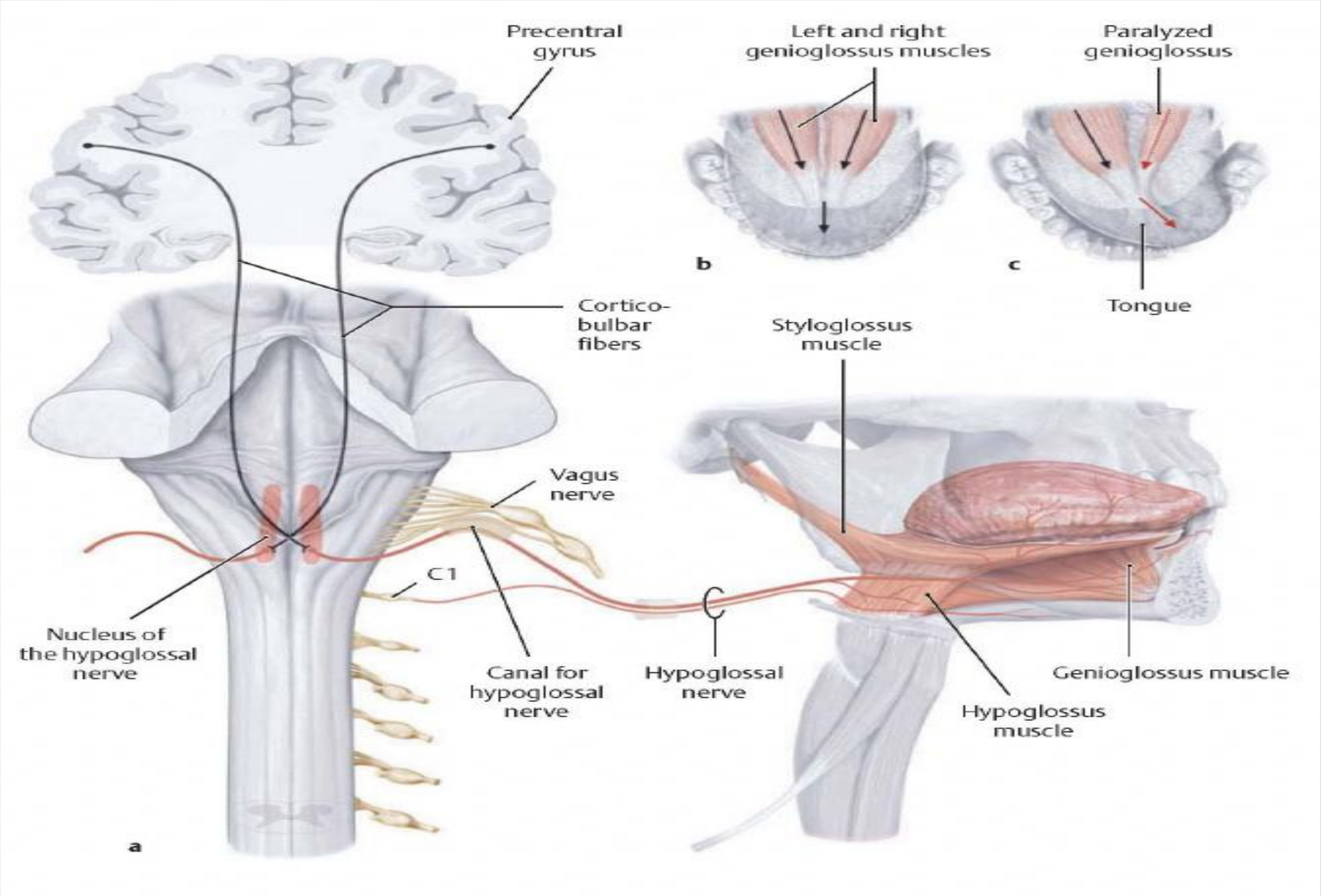
- Clinical/Osteopathic Considerations
  - As with both the cranial and spinal portions of the nerve, the jugular foramen is a common culprit as well as its association with the occipitomastoid suture and occipital condyles.
  - The spinal portion has the additional involvement of the cervical spine intervertebral foramina from C2 $\pm$ C6, foramen magnum issues as well as what is noted above. Due to very firm attachment of the sternocleidomastoid muscle, temporal bone dysfunction is probable.
  - Newborns with significant trauma can present with plagiocephaly (which has cranial base distortions) and torticollis. Neuropathy is common with compromise of the occipital condylar base

CN XII: HYPOGLOSSAL NERVE

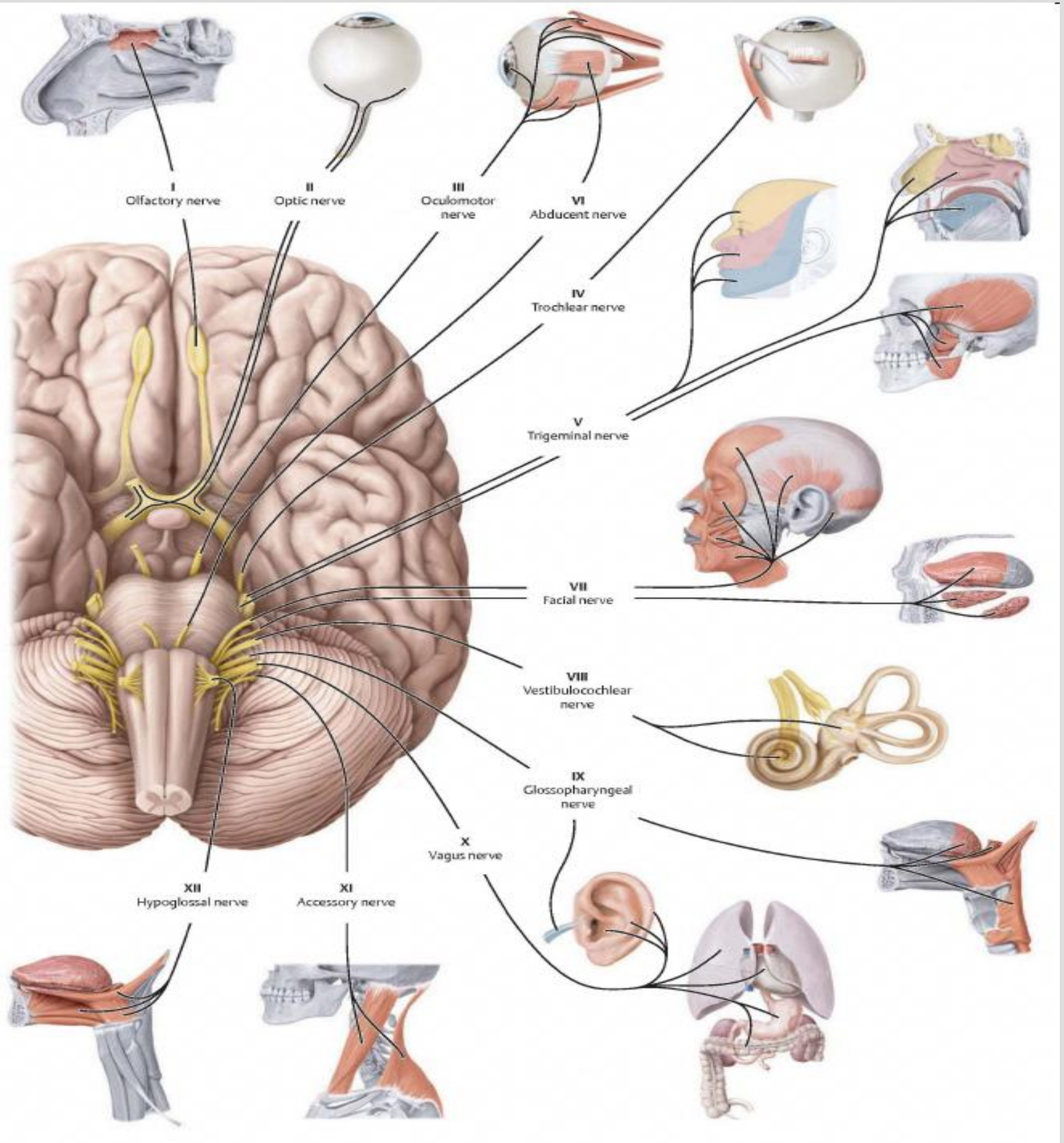
- *Origin:*
  - *Nucleus of the hypoglossal nerve*
- *Fiber Types*
  - *Somatic efferent*
    - *Muscles of the tongue*

- Path
  - Proceeds from the medulla oblongata through the hypoglossal canal that lies directly beneath the occipital condyles. After exiting ventral roots of C1/C2 travel with it for a short period then disembarks on their own journey.
  - Then enters the root of the tongue above the hyoid bone to innervate all of the tongue muscles with the exception of the palatoglossus (CN X).
- Clinical Osteopathic Considerations
  - Due to its proximity to the occipital condyle (grooved in the newborn) it is exposed to forces at the cranial base especial at birth. Frequently the newborn will present with poor latch and suck as well as tongue thrusting behavior in childhood.

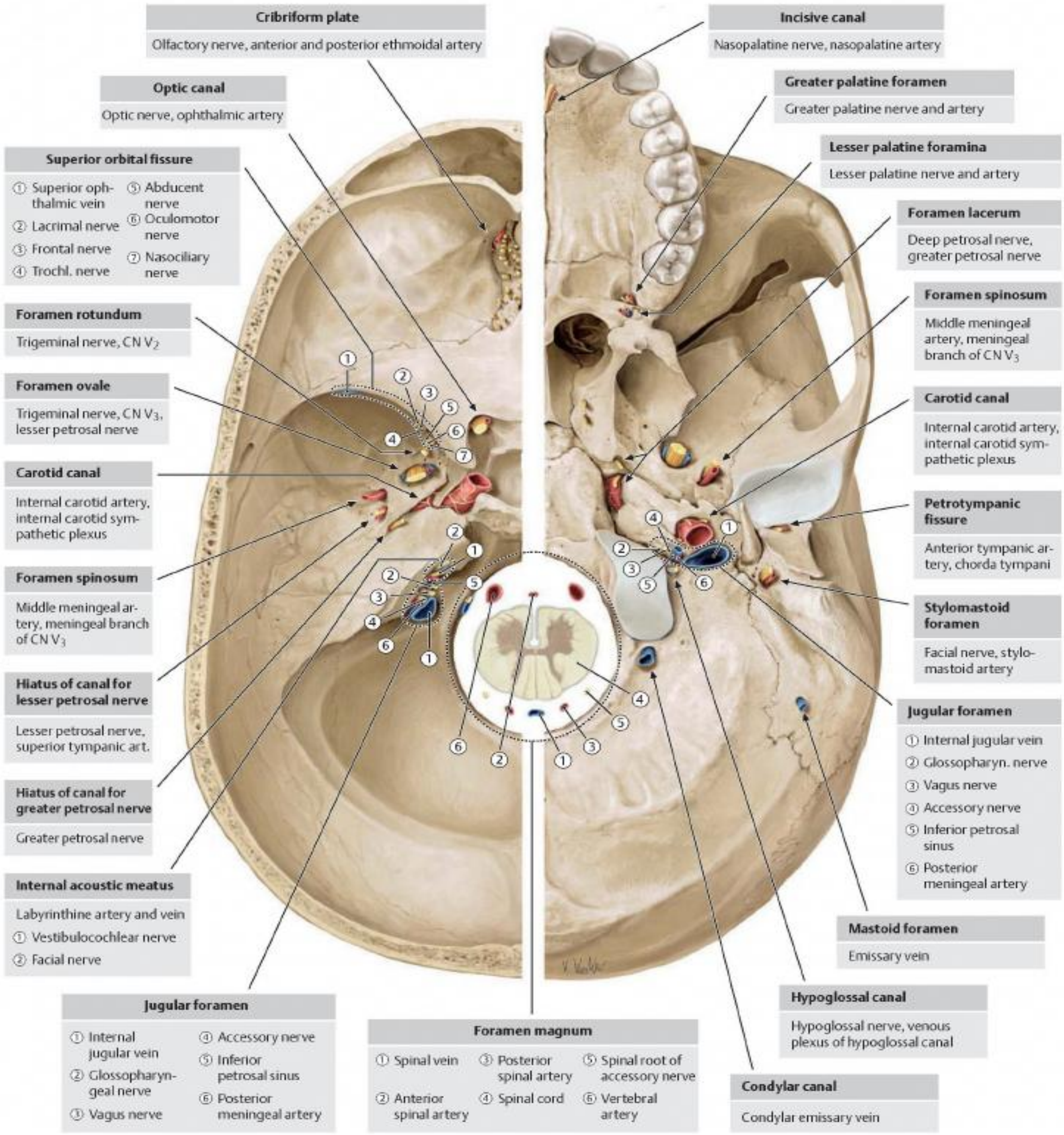
HYPOGLOSSAL NERVE (CN XII)



Cranial Nerve Overview



CN Foraminal Overview



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